Asymmetric price transmission in the Italian fuel market: Empirical evidence and methodological reflections

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Abstract

This thesis investigates asymmetric price transmission in the Italian retail fuel market, focusing on the "rockets and feathers" phenomenon whereby retail prices rise quickly when crude oil increases but fall more slowly when oil declines. Despite extensive international literature, empirical evidence on asymmetric adjustment remains mixed and methodologically sensitive, while comprehensive studies of the Italian market remain limited. Using weekly data from 2005 to 2024, the analysis covers multiple economic cycles and major disruptions including the global financial crisis and the COVID-19 pandemic. Retail gasoline and diesel pre-tax prices are matched with Brent crude oil prices, proxied primarily by 3-month futures to better reflect distributors' actual cost structures. After confirming that all price series are integrated of order one, long-run equilibrium relationships are identified through Engle–Granger cointegration analysis. A symmetric Vector Error Correction Model (VECM) is first estimated to establish baseline transmission dynamics, revealing significant price rigidity and slow convergence to equilibrium, with deviations corrected gradually over several months. The analysis then extends to nonlinear autoregressive distributed lag (NARDL) specifications to explicitly test for asymmetric responses to positive and negative oil price shocks. The results demonstrate incomplete and delayed pass-through from crude oil to retail prices, with asymmetric adjustment patterns consistent with the rockets-and-feathers hypothesis, particularly in the short run. Diesel prices exhibit slightly stronger and more persistent pass-through than gasoline. Overall, retail fuel prices in Italy respond sluggishly to cost decreases while reacting more promptly to increases, suggesting potential market inefficiencies or frictions. This analysis contributes to the limited empirical literature on the Italian fuel market and offers insights for competition policy and consumer welfare.

Contents

1 Introduction and Motivation **4**
- 1.1 Why Italy? Market Characteristics and Research Opportunity . . . 4
- 1.2 Research Contributions . . . 4
- 1.3 Research Questions and Analytical Framework . . . 5
- 1.4 Implications and Thesis Structure . . . 5

2 Literature Review **6**
- 2.1 Overview of Asymmetric Price Adjustment Literature . . . 6
- 2.2 Theoretical Foundations and Explanatory Mechanisms . . . 6
- 2.3 Methodological Evolution and Critical Sensitivity Issues . . . 6
- 2.4 Cross-Country Evidence and Market Heterogeneity . . . 7
- 2.5 Forward Prices versus Spot Prices: A Critical Methodological Choice . . . 8
- 2.6 The Italian Context: A Critical Research Gap . . . 8
- 2.7 Temporal Coverage and Structural Disruptions . . . 9
- 2.8 Summary and Research Positioning . . . 9

3 Data and methodology **10**
- 3.1 Data Sources and Sample Description . . . 10
  - 3.1.1 Data Quality and Market Coverage . . . 10
- 3.2 Variable Construction and Methodological Choices . . . 11
  - 3.2.1 Forward Prices vs. Spot Prices . . . 11
  - 3.2.2 Pre-tax vs. Post-tax Prices . . . 11
- 3.3 Descriptive Statistics and Preliminary Analysis . . . 11
  - 3.3.1 Price Evolution and Market Dynamics . . . 11
  - 3.3.2 Stationarity Analysis Through Autocorrelation Structure . . . 14
  - 3.3.3 Volatility Patterns in First Differences . . . 14
  - 3.3.4 Methodological Implications and Forward vs. Spot Price Validation . . . 17
- 3.4 Data Processing and Unit Root Testing . . . 17
  - 3.4.1 Missing Value Treatment . . . 17
  - 3.4.2 Unit Root Testing . . . 18
- 3.5 Cointegration and Causality Analysis . . . 19
  - 3.5.1 Optimal Lag Length Selection . . . 19
  - 3.5.2 Engle-Granger Cointegration Analysis . . . 21
- 3.6 Vector Error Correction Model (VECM) . . . 26
  - 3.6.1 Granger Causality Analysis . . . 33
  - 3.6.2 Impulse Response Functions . . . 36

4 Nonlinear Autoregressive Distributed Lag (NARDL) Analysis **43**
- 4.1 Motivation and Methodological Framework . . . 43
- 4.2 Cointegration Analysis: Bounds Test and Engle-Granger Verification . . . 45
  - 4.2.1 NARDL Bounds Test . . . 45
  - 4.2.2 Engle-Granger Test on ECT Stationarity . . . 46
  - 4.2.3 Synthesis and Implications . . . 47
- 4.3 Estimation Results and Model Performance . . . 48
  - 4.3.1 Long-Run Asymmetric Relationships . . . 48

4.3.2 Short-Run Dynamic Asymmetries 51
4.4 Formal Tests for Asymmetric Transmission 54
4.4.1 Wald Test Results 54
4.4.2 Why Short-Run Tests Lack Statistical Significance Despite Economic Asymmetry 55
4.5 Dynamic Multipliers and Adjustment Paths 56
4.6 Speed of Adjustment and Half-Lives 59
4.7 Model Diagnostics 60
4.8 Economic Interpretation and Policy Implications 62
4.9 Robustness, Limitations, and Directions for Future Research 63

5 Conclusion **65**

1 Introduction and Motivation

When crude oil prices plummet, consumers expect fuel prices to fall accordingly. Yet time and again, they witness prices that rise like rockets when oil costs increase, but drift down like feathers when oil becomes cheaper. This asymmetric adjustment pattern, known as the "rockets and feathers" effect, has puzzled economists and frustrated consumers for decades, generating a substantial body of research with mixed empirical results.

Despite extensive academic attention spanning multiple countries and time periods, empirical evidence on asymmetric fuel price adjustment remains inconclusive. Some studies find strong evidence of asymmetric behavior, while others document symmetric adjustment patterns. This inconsistency raises important questions about the prevalence and magnitude of such effects in different market contexts. The stakes are substantial, as asymmetric pricing behavior can impose significant welfare costs on consumers and may indicate market power abuse requiring regulatory intervention.

This study examines the presence and magnitude of asymmetric fuel price adjustment in the Italian market over an extended time period. Our primary objective is to determine whether the "rockets and feathers" effect characterizes the Italian fuel market, and to quantify the magnitude of any asymmetric patterns we identify.

1.1 Why Italy? Market Characteristics and Research Opportunity

The Italian fuel market offers a compelling case for asymmetric price analysis due to several distinctive features. The market exhibits oligopolistic concentration dominated by vertically integrated operators (ENI, Shell, IP, Q8), potentially facilitating coordinated pricing behavior. Italy also maintains one of Europe's highest fuel taxation levels—with taxes representing approximately 60% of the final price—introducing rigidities that may affect price adjustment patterns.

Beyond these structural characteristics, the Italian case presents a unique research opportunity through extensive temporal coverage (2005-2024) of high-frequency data. This extended dataset captures various economic cycles and market conditions, providing a comprehensive view of price transmission dynamics over nearly two decades. Unlike most European analyses, this study provides a long-span, high-frequency investigation of Italy's retail fuel markets using forward prices as cost proxies.

1.2 Research Contributions

This research addresses important gaps in the existing literature on asymmetric price adjustment:

First, long-term empirical evidence: Our 20-year dataset (2005-2024) represents one of the most comprehensive temporal analyses of Italian fuel price dynamics to date, enabling examination of long-term patterns and allowing us to distinguish between temporary adjustments and persistent market behavior.

Second, comprehensive market analysis: By examining the full sample period, we provide insights into the overall behavior of price transmission mechanisms in the Italian fuel market, capturing the aggregate effects of various market conditions and economic cycles.

Third, updated methodological approach: We employ nonlinear autoregressive distributed lag (NARDL) specifications to capture potential asymmetries in price adjustment, comparing results with a baseline symmetric error correction model. This framework allows for flexible dynamics while maintaining clear interpretation of short-run and long-run effects.

By providing a rigorous assessment over an extended time period, this study aims to contribute to understanding asymmetric adjustment patterns in the Italian fuel market.

1.3 Research Questions and Analytical Framework

Our analysis is guided by the following research questions:

1. Do Italian retail fuel prices adjust asymmetrically to changes in crude oil prices?
2. What are the short-run and long-run dynamics governing price transmission between crude oil and retail fuel prices?
3. What do the estimated adjustment dynamics reveal about the degree of price rigidity and market competitiveness in the Italian fuel sector?

Our analytical framework employs established error correction methodologies, beginning with a symmetric baseline specification and extending to nonlinear models that allow for asymmetric responses to cost increases and decreases. We use forward oil prices to better capture the actual costs faced by fuel distributors.

1.4 Implications and Thesis Structure

The results carry significant theoretical and practical implications. Theoretically, they contribute to understanding the mechanisms generating asymmetries in energy markets and their persistence over time. Practically, they provide evidence relevant for designing regulatory and monitoring policies, with potentially substantial implications for consumer welfare.

The remainder of this thesis proceeds as follows. Chapter 2 critically reviews the existing literature, covering theoretical foundations, methodological evolution, cross-country evidence, and the Italian context. Chapter 3 presents the dataset, variable construction, and econometric methodology, including unit root testing, cointegration analysis, and the symmetric VECM framework. Chapter 4 extends the analysis to the Nonlinear Autoregressive Distributed Lag (NARDL) model, testing for asymmetric price transmission and examining dynamic multipliers, adjustment speeds, and policy implications. Chapter 5 concludes with the key findings and directions for future research.

2 Literature Review

2.1 Overview of Asymmetric Price Adjustment Literature

The phenomenon of asymmetric price adjustment in fuel markets has captured the attention of researchers for decades, primarily due to its significant implications for consumer welfare and market competition. This asymmetry, commonly referred to as the "rockets and feathers" effect, describes the tendency of retail fuel prices to rise rapidly when crude oil prices increase, but to decline slowly when crude prices fall. The core hypothesis suggests that retail fuel prices respond faster and more completely to crude oil price increases than to decreases, imposing welfare costs on consumers and potentially indicating market power abuse.

Despite extensive academic attention spanning multiple countries and time periods, empirical evidence remains surprisingly mixed. Some studies find strong asymmetries while others detect symmetric adjustment, raising questions about whether these differences reflect true market heterogeneity or limitations in the analytical approaches employed.

2.2 Theoretical Foundations and Explanatory Mechanisms

The next section reviews the main theoretical mechanisms proposed to explain such asymmetries. The literature has identified three primary explanations, each receiving varying degrees of empirical support across different market contexts.

Oligopolistic Coordination: Gas stations may coordinate to maintain prices above competitive levels until sales drop below a threshold, signaling competitive pressure. When crude oil prices rise, stations must immediately raise prices to avoid losses, while decreases can be delayed until competitive forces intensify [Borenstein et al., 1997]. This "trigger sales" model receives mixed support—minimal at the wholesale level but stronger for retail coordination.

Production Lags and Inventory Constraints: This mechanism has garnered the most consistent empirical support. Refineries require 2-4 weeks for production adjustments, creating inventory imbalances that affect pricing dynamics. When crude prices rise, companies can quickly raise prices and build inventory; when prices should fall, existing supplies constrain downward adjustment [Borenstein et al., 1997]. The significant asymmetry in spot fuel prices directly supports this theory, as spot markets are highly competitive with low search costs.

Consumer Search Problems: During oil price volatility, consumers attribute price increases to market-wide changes rather than station-specific strategies, reducing shopping incentives and temporarily increasing retailers' pricing power. This finds support primarily at the retail level, where information asymmetries are most pronounced [Borenstein et al., 1997].

2.3 Methodological Evolution and Critical Sensitivity Issues

We then discuss how methodological innovations have evolved to address these challenges. The econometric approaches used to identify asymmetric price adjustment have evolved considerably, yet methodological choice has emerged as a critical determinant of empirical conclusions. Early studies relied on simple distributed lag models, but researchers recognized their limitations and moved toward more sophisticated frameworks incorporating long-run equilibrium constraints through error correction mechanisms.

Traditional Approaches: The partial adjustment model with threshold autoregressive error correction (TAR-ECM) techniques became the preferred methodology. These models offer

endogenous threshold estimation rather than fixing thresholds at zero, and have been advocated as particularly robust for identifying asymmetries [Bragoudakis and Sideris, 2021]. More recently, Non-linear Autoregressive Distributed Lag (NARDL) models have gained popularity, decomposing oil price changes into positive and negative partial sum processes while accommodating mixed orders of integration. Crucially, NARDL addresses methodological sensitivity by avoiding threshold-based regime identification, instead modeling asymmetries directly through cumulative positive and negative price change components.

The Methodological Sensitivity Problem: However, Martín-Moreno et al. [2019] demonstrate a critical issue: sophisticated TAR-ECM models may fail to capture asymmetries that more flexible approaches can identify. This finding is corroborated by Grasso and Manera [2007]—one of the few systematic studies including Italian data—who find that "the type of market and the number of countries characterized by asymmetry vary across models," with ultimately inconclusive results across different specifications.

The methodological comparison by Martín-Moreno et al. [2019] provides compelling evidence of this sensitivity. Their analysis of European fuel markets using both TAR-ECM and Markov-switching approaches reveals that "the M-S model is able to unveil asymmetric behaviours that the TAR-ECM model did not identify." Specifically, while TAR-ECM found no evidence of price asymmetry in gasoline markets, Markov-switching methodology identified short-run asymmetries consistent with the rockets-and-feathers hypothesis for the UK, France, and Italy, and long-run asymmetries for all countries analyzed. Alternative frameworks such as Markov-switching models capture latent structural regimes through endogenously determined states rather than fixed thresholds, offering greater flexibility but at the cost of substantial parameterization complexity and reduced interpretability, particularly when applied to long, high-frequency datasets. This sensitivity to model specification motivates the present study's adoption of the NARDL framework, which represents a middle ground: it preserves the interpretability of error correction frameworks while avoiding the rigidity of TAR-ECM threshold-based regime switches and the computational burden of Markov-switching estimation, accommodating asymmetric adjustment within a unified cointegrating structure that has gained substantial traction in recent empirical work.

2.4 Cross-Country Evidence and Market Heterogeneity

The international evidence presents a complex picture that varies considerably across countries, time periods, and methodological approaches. Early comprehensive evidence from the United States confirmed asymmetric adjustment patterns, with retail gasoline prices rising faster than they fall in response to crude oil changes [Borenstein et al., 1997]. However, more recent analyses present contrasting findings. Bumpass et al. [2015] find symmetric adjustment in both retail and wholesale gasoline markets using monthly data from 1976–2012, with retail prices showing faster adjustment (half-life of 2.25 months) compared to wholesale prices (4 months), suggesting limited long-run market power by retailers and wholesalers.

European evidence shows significant heterogeneity both across and within countries. Bettendorf et al. [2003] identify significant asymmetry in Dutch retail gasoline markets, but with critical methodological caveats—asymmetry depends on the chosen day of the week for data collection, with significant effects on Monday, Thursday, and Friday but symmetry on Tuesday and Wednesday, highlighting the surprising sensitivity of results to seemingly arbitrary analytical choices. Karagiannis et al. [2015] report that symmetry generally prevails in Euro-Area retail markets, yet individual country studies often contradict these aggregate findings. Bragoudakis and Sideris [2021] examine the Greek gasoline market over an extended period, finding asymmetric behavior

that varies across sub-periods, suggesting that temporal factors significantly influence asymmetric patterns.

Market structure, regulatory environments, and institutional factors further shape these dynamics. Hofstetter and Tovar [2008] provide unique evidence from Colombia's regulated market, where government reference prices create an institutional benchmark, finding asymmetry even in an environment of constantly rising costs. Alderighi and Baudino [2015] examine spatial pricing dynamics in Italian markets, finding limited price matching behavior and weak spatial propagation effects compared to more competitive markets, with price propagation following commuter paths rather than geographical proximity and diesel markets showing more reactive behavior than gasoline due to different consumer characteristics.

Recent cross-country analysis by Rehman et al. [2025], using daily data from 2009–2022 across eight developed economies, reveals significant heterogeneity: Germany, Italy, and the United States exhibit significant long-run asymmetries with slow convergence to equilibrium, while Australia, Canada, Japan, the UK, and France show long-run symmetry with proportional pass-through patterns, with adjustment speeds ranging from 10 days in Japan and Canada to prolonged periods in Germany, Italy, and the US.

2.5 Forward Prices versus Spot Prices: A Critical Methodological Choice

An important methodological consideration concerns the choice of crude oil benchmark. Martín-Moreno et al. [2019] advocate for using forward prices instead of spot prices, arguing that "fuel dealers use forward contracts to buy crude oil." This choice reflects actual market practices more accurately than spot price specifications commonly employed in earlier studies.

The rationale is compelling: retail fuel distributors typically purchase crude oil through forward contracts rather than spot markets, making forward prices more representative of their actual cost structure. Additionally, Grasso and Manera [2007] note that 66.7% of studies supporting asymmetric price behavior employ net-of-tax gasoline prices, as "asymmetries emerge more easily once the fiscal veil is removed" since taxes are beyond retailers' control.

Our approach follows this recommendation by utilizing 3-month forward prices, which better capture the actual costs faced by fuel distributors and eliminate the noise associated with near-month contracts that may not reflect underlying supply and demand fundamentals.

2.6 The Italian Context: A Critical Research Gap

Despite the extensive international literature on asymmetric fuel price adjustment, studies specifically focused on the Italian market present important limitations in scope and temporal coverage. This represents a notable research gap given Italy's distinctive market characteristics and economic importance within the European energy landscape.

Existing dedicated analyses of Italian fuel markets often suffer from constraints in temporal or geographic coverage. Alderighi and Baudino [2015] provide recent Italy-specific analysis, but their study focuses on a single medium-sized city (Cuneo) covering seven months of data from 2011. While their spatial analysis offers valuable insights into local competitive dynamics—finding limited price matching (approximately 30% matching frequency) and spatial dependence up to 1.1 km—the geographic and temporal scope limits broader market-level conclusions.

More recent contributions include Italy as part of broader cross-country studies—such as Martín-Moreno et al. [2019] and Rehman et al. [2025]—which necessarily sacrifice country-specific

institutional details for sample breadth. While these studies provide valuable comparative perspectives, they cannot capture the nuanced dynamics of Italy's specific market structure and regulatory environment. For instance, Rehman et al. [2025] include Italy in their eight-country analysis and find significant long-run asymmetries with slow convergence to equilibrium, but their aggregate approach cannot explain the underlying mechanisms or how they relate to specific Italian market characteristics.

Italy's unique features warrant dedicated analysis as they may generate asymmetric patterns differing from other European markets. The market exhibits concentration among vertically integrated operators (ENI, Shell, IP, Q8), and Italy maintains one of Europe's highest fuel taxation levels, with taxes representing approximately 60% of the final price. Italy's regulatory environment and enforcement mechanisms—including periodic investigations by the Competition Authority (AGCM) into fuel pricing practices—further distinguish it from comparable European markets. While the present study does not explicitly model these institutional variables, focusing exclusively on Italy over a long time horizon avoids the cross-country aggregation bias that arises when heterogeneous markets are pooled under homogeneous slope assumptions. The estimated transmission dynamics are therefore conditional on Italy's institutional environment and reflect its historically observed transmission mechanism, rather than an average across structurally diverse markets.

2.7 Temporal Coverage and Structural Disruptions

Extended temporal coverage has emerged as crucial for distinguishing between temporary adjustments and persistent market inefficiencies. The 2005-2024 period captures multiple economic cycles and, most importantly, the extraordinary disruption of the COVID-19 pandemic— representing an exogenous shock of unprecedented magnitude characterized by extreme oil price volatility (including historic negative oil prices in April 2020), lockdown measures that radically altered consumption patterns, and unprecedented government interventions in the energy sector.

This extended temporal dimension allows examination of whether asymmetric patterns persist across major structural disruptions or instead represent temporary phenomena sensitive to specific economic environments.

2.8 Summary and Research Positioning

The literature review reveals three critical findings that motivate our research design:

First, methodological sensitivity represents a fundamental challenge in this literature. The choice of econometric approach can significantly affect conclusions about the presence and magnitude of asymmetric pricing behavior, as demonstrated by contrasting results across different specifications in the existing literature.

Second, temporal coverage matters significantly for assessing the persistence of asymmetric patterns. Extended time series allow researchers to distinguish between temporary adjustments and persistent market inefficiencies, while the inclusion of major structural disruptions provides insight into whether asymmetries represent stable market characteristics.

Third, there exists a critical gap in comprehensive, recent analysis of the Italian market. While Italy features prominently in cross-country studies, no recent research provides the detailed, extended temporal analysis necessary to understand the evolution of price transmission mechanisms in this distinctive market characterized by unique institutional features and market structure. Our 20-year dataset (2005-2024) therefore represents one of the most comprehensive temporal analyses of Italian fuel price dynamics to date.

In summary, the existing evidence on fuel price asymmetry remains inconclusive, partly due to methodological considerations and partly to market-specific factors. Our study contributes by applying a unified econometric framework combining symmetric and asymmetric error correction specifications to the Italian market over an extended period that encompasses major economic disruptions. This approach enables us to assess whether asymmetric price transmission represents a persistent feature of the Italian fuel market or a phenomenon sensitive to specific market conditions.

3 Data and methodology

3.1 Data Sources and Sample Description

This study employs weekly time series data spanning from 2005 to 2024, providing comprehensive 20-year coverage of Italian fuel price dynamics. The extended temporal window enables examination of long-term price transmission patterns while capturing major structural disruptions, including the 2008 financial crisis and the COVID-19 pandemic.

Fuel Price Data: Italian retail gasoline and diesel prices are sourced from the European Commission's Weekly Oil Bulletin, which provides harmonized fuel price data across EU member states. Prices are expressed in euros per 1,000 liters and represent pre-tax retail prices, following Grasso and Manera [2007], who demonstrate that asymmetries emerge more clearly once the fiscal component is removed, as taxes are beyond retailers' control and may mask underlying price transmission mechanisms.

Crude Oil Data: Crude oil prices are obtained from the LSEG (London Stock Exchange Group) Database. Our primary analysis employs the 3rd continuous futures contract as a proxy for 3-month forward prices, which maintains a maturity close to three months and provides superior liquidity without the noise associated with near-month contracts. Additionally, we include spot prices to provide comparative evidence on which benchmark better captures price transmission to retail fuel markets.

Exchange Rate Data: To maintain temporal consistency and eliminate currency risk, oil prices are converted from US dollars per barrel to euros per barrel using exchange rates sourced from the LSEG Database. For futures-based oil prices, we employ corresponding 3-month forward exchange rates to maintain maturity consistency, while spot oil prices are converted using contemporaneous spot exchange rates.

Data Frequency: Weekly data frequency represents an optimal balance between capturing short-term price dynamics and avoiding the excessive noise of daily observations, while providing substantially more observations than monthly data for robust statistical inference. The final dataset comprises approximately 1,000 weekly observations per series.

3.1.1 Data Quality and Market Coverage

The Italian fuel price data are collected by the Ministry of Economic Development using weighted averages based on actual sales volumes, providing more accurate market representation than simple arithmetic means. The collection process encompasses all 10 large oil companies operating in Italy, supplemented by independent traders and approximately 30 supermarket chains, achieving market coverage above 95% for each product category. This extensive coverage captures the full spectrum of Italian retail fuel distribution, from vertically integrated major operators (ENI, Shell, IP, Q8) to independent retailers and supermarket chains.

Importantly, reported prices incorporate various discount schemes (fidelity discounts, supermarket client discounts, seasonal offers), capturing actual transaction prices rather than posted prices—addressing a common limitation in fuel price transmission studies. The Ministry conducts monthly surveys to determine market shares based on quantities sold, ensuring that weighted averages accurately reflect the relative importance of different distribution channels.

3.2 Variable Construction and Methodological Choices

3.2.1 Forward Prices vs. Spot Prices

The choice between forward and spot prices represents a crucial methodological decision. While spot prices are commonly employed in the literature, theoretical considerations suggest that forward prices may better represent actual distributor cost structures, as fuel distributors typically purchase crude oil through forward contracts rather than spot markets.

Following Martín-Moreno et al. [2019], who demonstrate that forward prices exhibit stronger co-movement with retail fuel prices, our primary analysis employs the 3rd continuous futures contract as a proxy for 3-month forward prices. This choice reflects actual market practices more accurately than spot price specifications. The use of the 3rd continuous futures contract addresses the practical constraint that comprehensive forward price databases are not readily available, while futures contracts with similar maturities provide highly correlated price movements and are standard in energy economics literature.

We include spot prices in our preliminary analysis to provide robustness checks on our main findings.

3.2.2 Pre-tax vs. Post-tax Prices

The analysis focuses on pre-tax retail prices to isolate the price transmission mechanism between crude oil costs and retail margins. Italy's exceptionally high fuel tax burden—with taxes representing approximately 60% of final consumer prices—could potentially mask underlying asymmetric adjustment patterns. By analyzing pre-tax prices, we can better identify whether asymmetries stem from market power or competitive dynamics rather than tax-related rigidities.

3.3 Descriptive Statistics and Preliminary Analysis

This section presents descriptive statistics and preliminary examination of our key variables, providing insights into distributional properties, stationarity characteristics, and volatility patterns.

3.3.1 Price Evolution and Market Dynamics

Figure 1 presents the evolution of Italian fuel prices over our 20-year sample period, while Figure 2 displays both spot and 3-month futures Brent crude oil prices.

The complementary analysis of crude oil price evolution is presented in Figure 2, which displays both spot and 3-month futures Brent crude oil prices over the same period. The separate presentation allows for proper scale visualization while facilitating direct comparison of price transmission patterns.

Table 1 summarizes the quantitative characteristics of price levels.

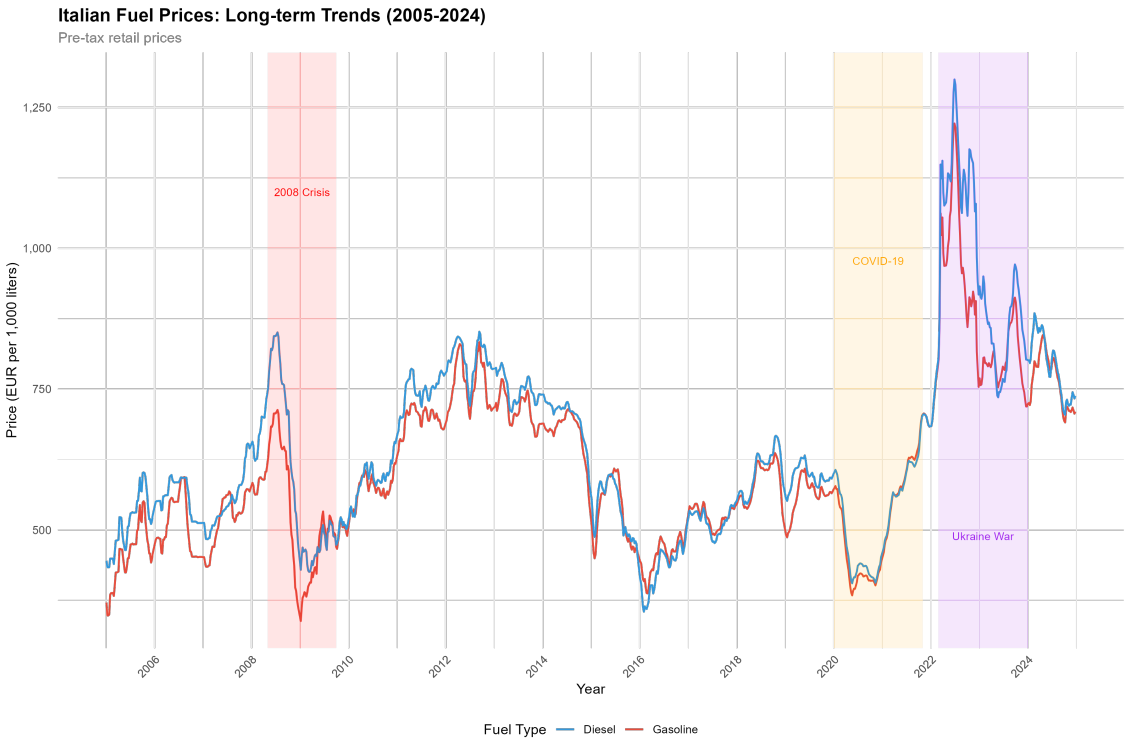


Figure 1: Evolution of Italian Fuel Prices (2005-2024)

Note: The figure displays weekly time series of pre-tax Italian gasoline and diesel prices (EUR per 1,000 liters). Key periods include the 2008 financial crisis, 2014-2016 oil price collapse, and the unprecedented volatility during the COVID-19 pandemic (2020-2021).

Table 1: Descriptive Statistics for Italian Fuel Price Series (Levels, 2005-2024)

Series	N	Mean	Median	Std. Dev.	Min	Max	Skewness	Kurtosis	CV
Gasoline (levels)	993	609.85	577.01	144.63	338.20	1221.58	0.88	4.20	0.24
Diesel (levels)	993	643.19	599.38	164.71	354.85	1299.67	1.06	4.44	0.26
Brent 3M (levels)	993	62.03	59.30	16.54	25.00	110.03	0.23	2.23	0.27
Brent SPOT (levels)	993	61.96	58.92	18.08	13.16	121.14	0.28	2.61	0.29

Note: Fuel prices are expressed in euros per 1,000 liters (pre-tax); crude oil prices in USD per barrel. CV denotes coefficient of variation.

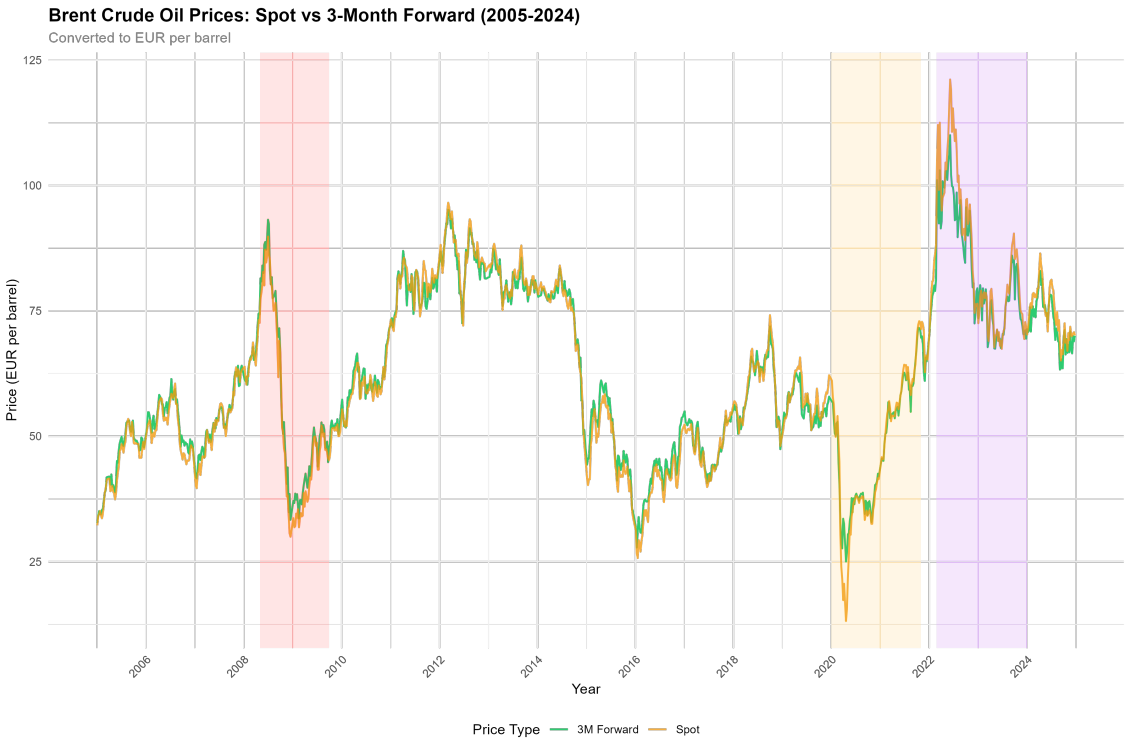


Figure 2: Evolution of Brent Crude Oil Prices: Spot vs. 3-Month Futures (2005-2024)
Note: The figure shows weekly time series of Brent crude oil spot prices and 3-month futures contracts (USD per barrel). The close co-movement between spot and futures prices is evident, with futures exhibiting slightly less volatility during extreme market conditions, particularly during the COVID-19 crisis of 2020.

Visual inspection suggests non-stationary behavior typical of commodity price series, with all series exhibiting high persistence and structural breaks during the 2008 financial crisis and COVID-19 pandemic. The substantial price variability—with gasoline prices varying nearly four-fold from €338 to €1,222 per 1,000 liters—reflects the extraordinary market disruptions captured in our sample.

3.3.2 Stationarity Analysis Through Autocorrelation Structure

Figure 3 presents autocorrelation function (ACF) and partial autocorrelation function (PACF) analysis for all price series in levels.

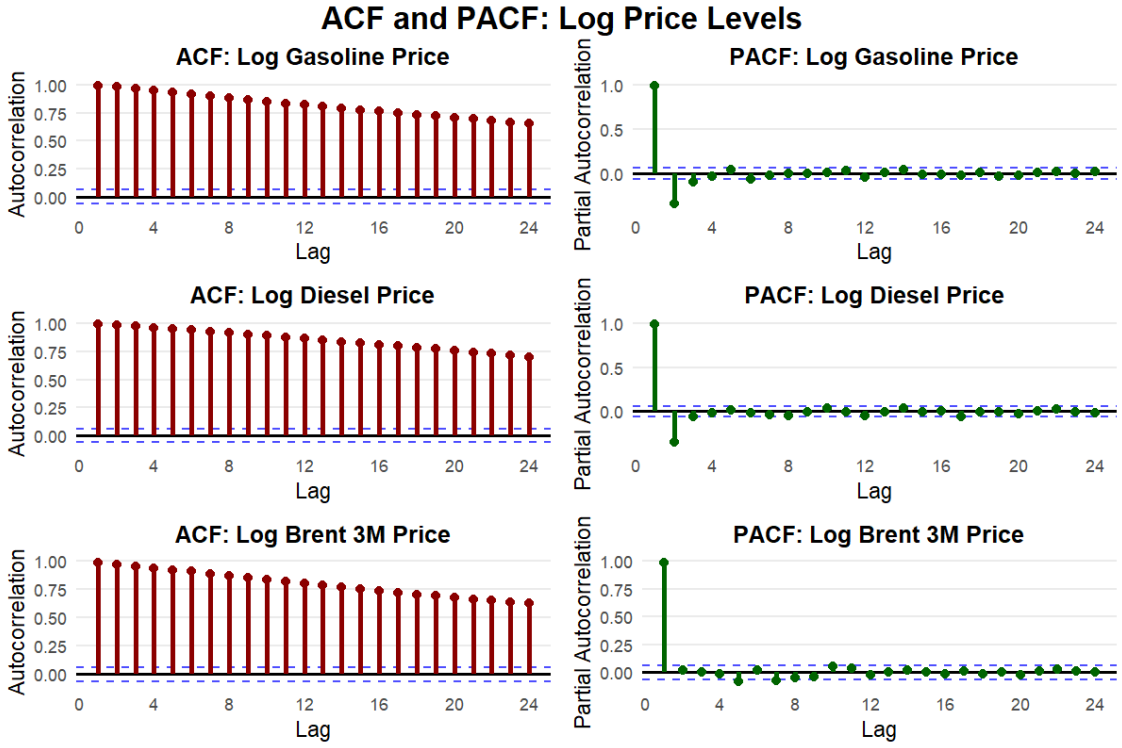


Figure 3: Autocorrelation and Partial Autocorrelation Functions for Price Series in Levels

Note: The figure displays ACF (left panels) and PACF (right panels) for gasoline, diesel, and Brent 3M futures prices in logarithmic levels. The persistent high autocorrelation coefficients in the ACF and the large initial spike in the PACF provide clear evidence of non-stationarity in all three series. The dashed lines represent 95% confidence intervals under the null hypothesis of white noise.

The ACF patterns exhibit characteristic features of non-stationary series: autocorrelation coefficients remain persistently high (exceeding 0.8) even at extended lags of 20-30 weeks, far above the 95% confidence intervals. The PACF analysis reinforces this conclusion, with all three series exhibiting large first-order partial autocorrelations (exceeding 0.9) followed by much smaller higher-order coefficients—textbook evidence of unit root processes. These patterns provide strong preliminary evidence of non-stationarity, which we formally test in Section 3.4.

3.3.3 Volatility Patterns in First Differences

Given the confirmed non-stationarity of price levels, we examine first differences (log changes) that form the core variables for our econometric models. Figures 4 and 5 display weekly percentage changes.

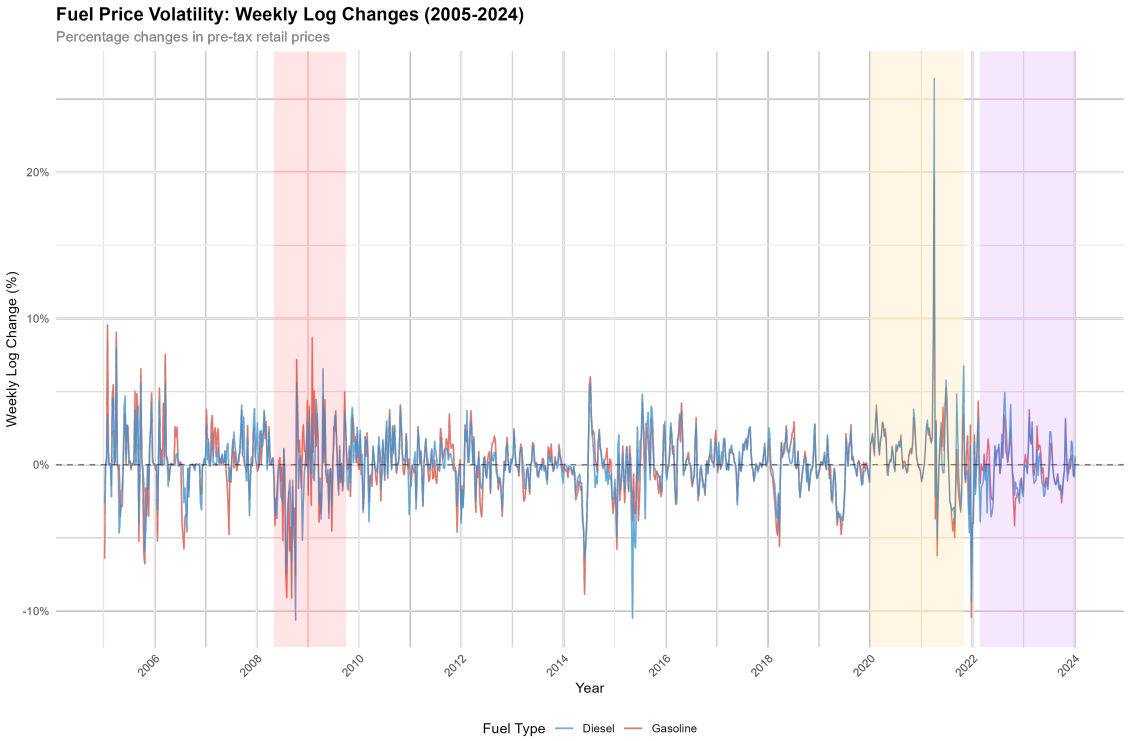


Figure 4: Weekly Fuel Price Changes and Volatility Clustering (2005-2024)

Note: The figure shows weekly log differences (percentage changes) for Italian gasoline and diesel prices. Volatility clustering is particularly evident during the 2008 financial crisis, 2014-2016 oil price collapse, and COVID-19 pandemic. The volatility patterns provide visual evidence of the smoothing mechanisms operating in retail fuel markets compared to crude oil inputs.

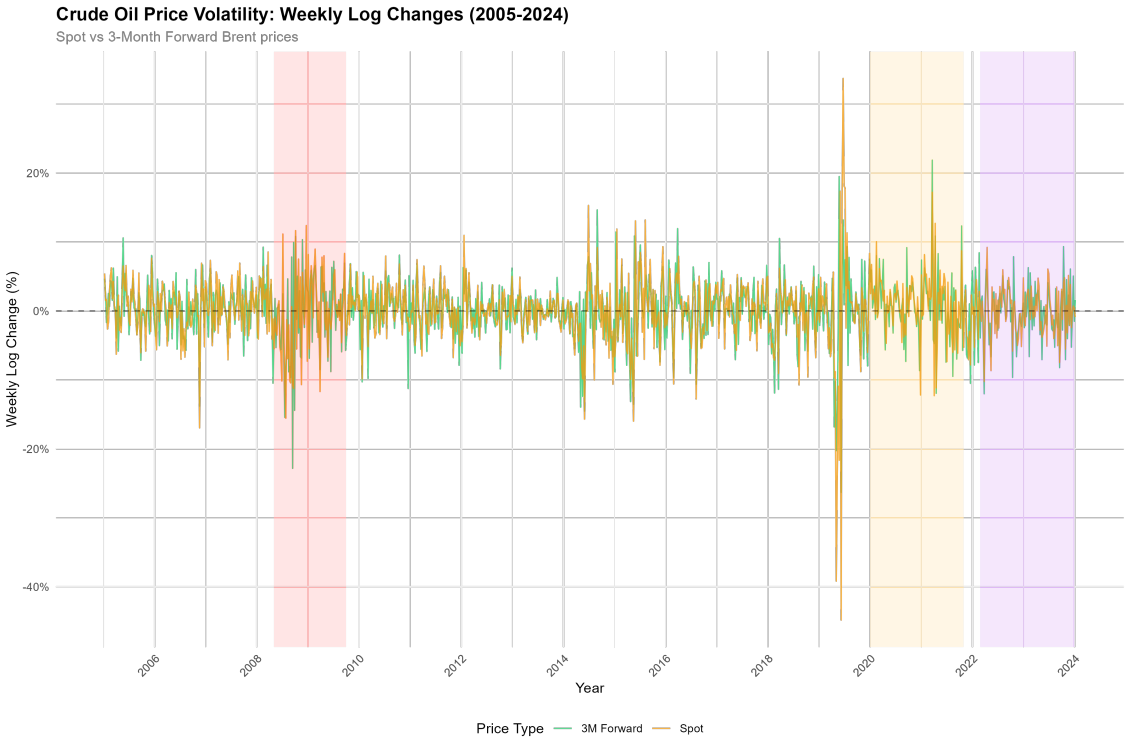


Figure 5: Weekly Oil Price Changes and Volatility Clustering (2005-2024)

Note: The figure shows weekly log differences (percentage changes) for Brent crude oil futures and spot prices. Crude oil exhibits substantially higher volatility than retail fuel prices, with occasional extreme movements exceeding 20-30% during the COVID-19 period. The comparison between spot and futures prices reveals that futures contracts provide a cleaner signal with marginally lower extreme volatility.

The comparative analysis of Figures 4 and 5 reveals the fundamental asymmetric adjustment patterns that motivate our entire analysis. Visual inspection suggests that positive oil price shocks (upward spikes in Figure 5) appear to be transmitted more rapidly and completely to retail fuel prices (Figure 4), while negative shocks (downward movements) exhibit more gradual transmission—precisely the "rockets and feathers" effect we seek to quantify econometrically.

Table 2 quantifies the volatility patterns.

Table 2: Descriptive Statistics for Italian Fuel Price Series (First Differences in Logarithms, 2005-2024)

Series	N	Mean	Median	Std. Dev.	Min	Max	Skewness	Kurtosis	CV
Δ Log Gasoline	992	0.0007	0.0003	0.0219	-0.1046	0.1967	0.47	11.73	33.58
Δ Log Diesel	992	0.0005	0.0000	0.0213	-0.1062	0.2641	1.61	28.07	41.95
Δ Log Brent 3M	992	0.0008	0.0036	0.0445	-0.2631	0.2187	-0.58	6.66	57.98
Δ Log Brent SPOT	992	0.0008	0.0019	0.0485	-0.4482	0.3371	-1.16	18.01	61.94

Note: All variables are expressed as weekly log differences, representing percentage changes. CV denotes coefficient of variation.

The volatility hierarchy is clear: crude oil exhibits roughly double the volatility of retail fuel prices (4.45-4.85% vs 2.13-2.19%), reflecting smoothing mechanisms operating through inventory management, menu costs, and competitive dynamics in retail markets. This volatility differential provides the theoretical foundation for potential asymmetric adjustment—if retail prices were perfectly flexible, they should exhibit volatility similar to their input costs.

Visual inspection of Figures 4 and 5 suggests that positive oil price shocks appear to transmit more rapidly to retail fuel prices than negative shocks—the "rockets and feathers" effect we seek to quantify econometrically. The comparison between spot and futures prices reveals that futures contracts exhibit moderately lower extreme values (maximum absolute changes of 26% vs 45%), though both series display similar volatility clustering patterns during crisis periods.

3.3.4 Methodological Implications and Forward vs. Spot Price Validation

The descriptive analysis reveals several key features of our data: (1) all price series exhibit high persistence suggesting potential non-stationarity, which we formally test in the next section; (2) crude oil exhibits substantially higher volatility than retail fuel prices, consistent with smoothing mechanisms in retail markets; (3) visual inspection suggests potential asymmetric adjustment patterns, with retail prices appearing to respond more rapidly to positive oil shocks than negative ones—the central hypothesis we test in subsequent sections.

3.4 Data Processing and Unit Root Testing

3.4.1 Missing Value Treatment

The Weekly Oil Bulletin data contains approximately 50 missing observations over the 20-year period (less than 5% of total observations), primarily reflecting occasional gaps in weekly reporting. Missing observations are excluded through listwise deletion rather than interpolation, preserving the authentic volatility characteristics of the data while maintaining time series integrity. The limited extent of missing data ensures this approach does not materially compromise statistical power or temporal coverage, and robustness checks confirm that gaps are randomly distributed rather than concentrated in specific periods or market conditions.

3.4.2 Unit Root Testing

To establish the integration properties of our price series, we employ a comprehensive battery of unit root tests: Augmented Dickey-Fuller (ADF), Phillips-Perron (PP), KPSS, and Zivot-Andrews. The ADF and PP tests examine the null hypothesis of a unit root against stationarity, while KPSS reverses the null to test stationarity against non-stationarity. The Zivot-Andrews test extends the ADF framework by allowing for an endogenously determined structural break, which is particularly relevant given our sample encompasses the 2008 financial crisis and COVID-19 pandemic.

Following standard practice for commodity prices [Hamilton, 2003, Pindyck, 1999], we employ constant-only specifications in all tests, as energy prices are characterized by stochastic trends rather than deterministic time trends. This approach is consistent with recent studies on oil price dynamics [Baumeister and Kilian, 2016] and European fuel markets [Perdiguero-Garcia, 2013]. Optimal lag lengths are selected using the Akaike Information Criterion to ensure white noise residuals.

Table 3 presents the unit root test results for all three price series using intercept-only specifications.

Table 3: Unit Root Test Results

	ADF	PP	KPSS	Zivot-Andrews
	H_0: UR	H_0: UR	H_0: ST	H_0: UR (break)
Gasoline Price	-2.886	-3.111	0.883	-4.333
Diesel Price	-2.474	-2.315	1.604	-3.131
Brent Crude Oil (3M)	-3.064	-2.816	2.342	-3.340
Critical Values				
1%	-3.435	-3.435	0.574	-5.340
5%	-2.868	-2.868	0.463	-4.800
10%	-2.570	-2.570	0.347	-4.580

Notes: ADF and PP include drift. KPSS tests level stationarity. Zivot-Andrews allows for an intercept break. Lag length selected via AIC.

Empirical Results The unit root test results provide convergent evidence for non-stationarity across all price series, though with varying strength depending on the test employed.

The ADF and Phillips-Perron tests yield mixed evidence. While neither test formally rejects the unit root null hypothesis at conventional significance levels, several statistics approach borderline significance. Gasoline prices show the strongest evidence: the ADF statistic of -2.886 nearly reaches the 5% critical value of -2.868, while the PP statistic of -3.111 exceeds it marginally. Brent crude oil also exhibits relatively strong evidence with an ADF statistic of -3.064. However, diesel prices show weaker evidence against the unit root hypothesis. Overall, while suggestive, these borderline results remain insufficient for formal rejection of non-stationarity.

The Zivot-Andrews tests, which endogenously identify potential structural breaks, provide robust confirmation of the unit root hypothesis. Test statistics range from -3.131 (diesel) to -4.333 (gasoline), all falling substantially short of even the 10% critical value of -4.580. This finding is particularly informative: even when explicitly allowing for endogenous regime changes—a feature designed to increase test power in the presence of structural instability—the data consistently support non-stationarity across all series.

The KPSS tests provide the strongest evidence for non-stationarity. With the null hypothesis reversed to stationarity, all three series decisively reject this hypothesis at the 1% significance level. The KPSS statistics (0.883 for gasoline, 1.604 for diesel, and 2.342 for Brent crude oil) substantially exceed the 1% critical value of 0.574. The magnitude of these statistics, particularly for diesel and crude oil, indicates strong non-stationary behavior and provides robust evidence against stationarity.

Integration Order Conclusion: The weight of evidence across all four tests converges toward the same conclusion. While ADF and PP tests show borderline evidence, the Zivot-Andrews tests—robust to structural breaks—confirm the unit root hypothesis, and the KPSS tests decisively reject stationarity. Combined with the theoretical expectation that energy prices typically exhibit I(1) behavior [Hamilton, 2003, Pindyck, 1999], we conclude that all three price series—gasoline, diesel, and Brent crude oil (3-month futures)—are integrated of order one. This I(1) property provides the necessary foundation for valid cointegration analysis and justifies our subsequent Error Correction Model framework for examining asymmetric price adjustment dynamics.

3.5 Cointegration and Causality Analysis

Having established the I(1) nature of our price series through comprehensive unit root testing, we examine the long-run equilibrium relationships between crude oil and retail fuel prices in the Italian market. As discussed in Section 3.2.1, 3-month forward prices are used as the primary crude oil proxy, given their closer alignment with distributors' actual cost structures. This analysis identifies stable long-run relationships that will form the foundation for the asymmetric adjustment modeling developed in Chapter 4.

3.5.1 Optimal Lag Length Selection

The determination of appropriate lag length in Vector Error Correction Models represents a crucial specification choice that balances model adequacy against parametric parsimony. However, the appropriate framework for lag selection depends critically on the system's properties and the subsequent modeling strategy [Lütkepohl, 2005].

Methodological Approach: Bivariate Systems in First Differences Given that our subsequent analysis focuses on bivariate asymmetric adjustment models (oil-gasoline and oil-diesel separately), we conduct lag selection on two distinct bivariate VAR systems estimated in first differences rather than a single multivariate VAR in levels. This approach is motivated by three considerations.

First, our cointegration analysis employs the bivariate Engle-Granger methodology, establishing separate long-run relationships for each fuel type. The VECM specification naturally follows this bivariate structure, making it methodologically consistent to perform lag selection on the same system that will be estimated [Enders, 2015]. As Hamilton [1994] emphasizes, lag selection should be performed on the same specification used for subsequent inference to avoid pre-test bias.

Second, estimating VAR models in first differences is standard practice when the objective is to determine the lag structure for a VECM with known cointegrating relationships [Juselius, 2006]. The lag order p selected for the differenced VAR corresponds directly to the number of lagged difference terms in the VECM specification, with the error correction term capturing the additional long-run dynamics [Lütkepohl and Krätzig, 2004]

Third, separate bivariate systems allow for potential heterogeneity in adjustment dynamics between gasoline and diesel markets. Borenstein et al. [1997] and Lewis [2011] document that gasoline and diesel exhibit distinct adjustment patterns due to differences in demand elasticities, refining processes, and market structures. Imposing a common lag structure through a trivariate system would obscure these fuel-specific dynamics.

For each bivariate VAR model with p lags and $K = 2$ variables, the information criteria are defined as:

$$AIC(p) = \ln |\hat{\Sigma}_u(p)| + \frac{2pK^2}{T} \tag{1}$$

$$BIC(p) = \ln |\hat{\Sigma}_u(p)| + \frac{pK^2 \ln(T)}{T} \tag{2}$$

where $\hat{\Sigma}_u(p)$ denotes the estimated residual covariance matrix, T represents the sample size, and p is the lag order. The optimal lag length minimizes the chosen criterion: $p^* = \arg\min_p IC(p)$.

The key distinction between these criteria lies in their penalty terms: BIC imposes a more stringent penalty (proportional to $\ln(T)$) compared to AIC (constant penalty of 2), systematically favoring more parsimonious specifications. This difference becomes particularly relevant in finite samples, where AIC's lenient penalty can lead to overparameterization—a critical concern in energy market applications where excessive lags may capture spurious dynamics rather than genuine price transmission mechanisms [Kilian and Lütkepohl, 2017].

Table 4 presents the information criteria values across candidate lag lengths from 1 to 12 weeks for both the oil-gasoline and oil-diesel systems.

Table 4: VAR Lag Selection: AIC and BIC for Oil-Gasoline and Oil-Diesel Systems

Lag	AIC (Gasoline)	BIC (Gasoline)	AIC (Diesel)	BIC (Diesel)
1	-14.2989	-14.2690	-14.3164	-14.2865
2	-14.5047	-14.4549	-14.5558	-14.5060
3	-14.5307	**-14.4609**	-14.6025	-14.5327
4	-14.5396	-14.4498	-14.6258	**-14.5360**
5	-14.5370	-14.4272	-14.6262	-14.5165
6	-14.5394	-14.4097	-14.6288	-14.4991
7	-14.5433	-14.3937	-14.6346	-14.4849
8	-14.5421	-14.3725	-14.6367	-14.4671
9	**-14.5476**	-14.3580	**-14.6412**	-14.4517
10	-14.5462	-14.3367	-14.6362	-14.4267
11	-14.5389	-14.3095	-14.6303	-14.4009
12	-14.5401	-14.2907	-14.6323	-14.3829

Note: Bold values indicate the optimal lag selection for each criterion. VAR models estimated on first-differenced series.

Criterion Divergence and Selection Rationale The results reveal the characteristic divergence between AIC and BIC. For the gasoline system, AIC selects 9 lags while BIC selects 3 lags. For diesel, AIC identifies 9 lags as optimal while BIC selects 4 lags. This pattern is well-documented in the VAR literature and reflects the fundamental trade-off between model flexibility and statistical efficiency [Hannan and Quinn, 1979].

Several considerations support favoring BIC's more parsimonious specifications in our context. First, the gradual flattening of AIC values beyond 3-4 lags suggests diminishing marginal improvements in model fit—a classic sign of potential overparameterization. For gasoline, AIC improves by only 0.017 points (from -14.5307 to -14.5476) between lags 3 and 9; for diesel, the improvement is 0.039 points. Such marginal gains risk capturing noise rather than genuine adjustment dynamics [Lütkepohl, 2005].

Second, in the specific context of fuel price transmission, Acaravci et al. [2012] demonstrate that 2-4 weeks typically suffice to capture primary adjustment dynamics in European energy markets, as most retail price responses to crude oil shocks materialize within this window. Similarly, Grasso and Manera [2007] find that Italian fuel prices exhibit rapid initial adjustment followed by gradual convergence, with the bulk of the response occurring within the first month.

Third, from an economic perspective, a 3-4 week lag structure aligns well with the institutional features of Italian fuel markets. Retail fuel prices typically respond to crude oil cost changes within 2-3 weeks, reflecting the combination of: (i) wholesale contract adjustment periods (typically 1-2 weeks), (ii) inventory turnover rates in the distribution chain, and (iii) competitive pricing cycles among retailers [Grasso and Manera, 2007]. Lag structures extending to 9 weeks would capture dynamics well beyond the primary transmission mechanism, potentially confounding genuine price adjustment with unrelated market noise.

Final Specification We therefore adopt **3 lags** for both the gasoline and diesel VECM specifications, following the BIC criterion for gasoline and prioritizing consistency across fuel types. While BIC selects 4 lags for diesel, adopting a uniform 3-lag structure facilitates direct comparison of adjustment dynamics between the two fuels and ensures that our VECM estimations capture the essential short-run price transmission mechanisms without excessive parameterization. This choice optimally balances model adequacy with parametric parsimony, ensuring efficient estimation and robust inference in subsequent asymmetric adjustment analysis [Enders, 2015].

3.5.2 Engle-Granger Cointegration Analysis

Before formally testing for cointegration, we begin by visualizing the log-transformed price series to assess their long-run co-movement patterns. Figure 6 plots the logarithmic prices of Brent crude oil (3-month futures), gasoline, and diesel over the full sample period, with shaded areas highlighting the three major crisis periods that characterize our analysis window.

The visual inspection reveals several important features that motivate our cointegration analysis. First, all three price series exhibit strong co-movement throughout the sample period, with gasoline and diesel prices closely tracking movements in crude oil prices. This pattern is particularly evident during the major price swings associated with the 2008 financial crisis, the COVID-19 demand collapse, and the Ukraine war supply disruptions. Second, while the three series move together in the long run, their short-run dynamics reveal notable deviations from perfect co-movement, especially during crisis periods. These temporary divergences from equilibrium suggest the presence of adjustment mechanisms that warrant formal econometric investigation. Third, the vertical distance between the fuel price series and the oil price series remains relatively stable over time despite substantial price level variations, providing visual evidence consistent with a stable long-run elasticity relationship.

The analysis of long-term relationships between oil prices and fuel prices represents a crucial step in understanding the transmission mechanisms of price shocks in energy markets. Following

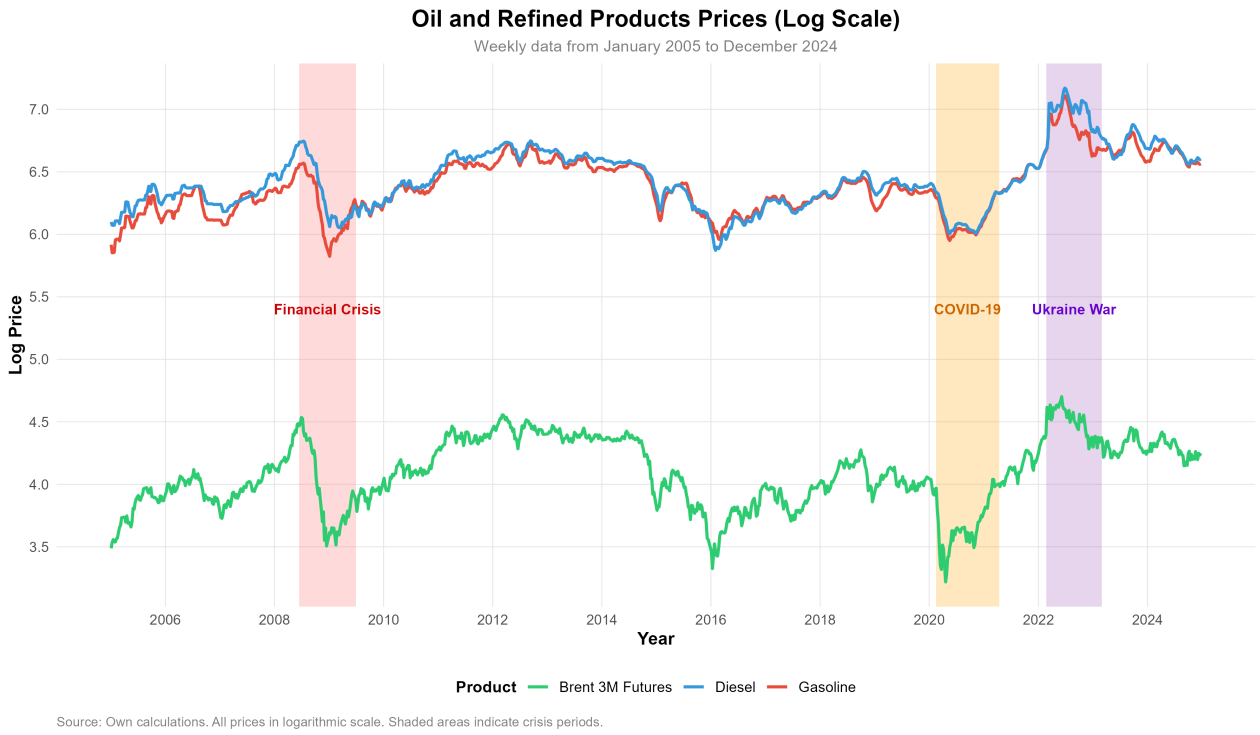


Figure 6: Log-Transformed Oil and Fuel Prices with Crisis Periods (2005-2024)

Note: All prices are in logarithmic scale. The green line represents Brent crude oil 3-month futures prices (EUR per barrel), while red and blue lines show Italian gasoline and diesel pre-tax retail prices (EUR per 1,000 liters), respectively. Shaded areas denote the 2008 financial crisis (red), COVID-19 pandemic (orange), and Ukraine war period (purple). The visual co-movement across all three series provides preliminary evidence of long-run equilibrium relationships.

the seminal work of Engle and Granger [1987], we employ the two-step Engle-Granger methodology to test for cointegration between Brent crude oil prices and Italian gasoline and diesel prices.

The Engle-Granger procedure consists of two distinct steps. In the **first step**, we estimate the long-run cointegrating regression using Ordinary Least Squares (OLS):

$$y_t = \alpha + \beta x_t + u_t \tag{3}$$

where $y_t = \ln(P_{fuel,t})$ represents the log of fuel prices, $x_t = \ln(P_{oil,t})$ is the log of oil prices, and u_t denotes the equilibrium error. Under cointegration, the residuals $\hat{u}_t = y_t - \hat{\alpha} - \hat{\beta} x_t$ must be stationary despite both y_t and x_t being I(1) processes [Stock, 1987].

In the **second step**, we test the stationarity of the estimated residuals using the Augmented Dickey-Fuller (ADF) test specifically designed for cointegration testing:

$$\Delta \hat{u}_t = \rho \hat{u}_{t-1} + \sum_{j=1}^{k} \gamma_j \Delta \hat{u}_{t-j} + \epsilon_t \tag{4}$$

where the null hypothesis $H_0 : \rho = 0$ (unit root, no cointegration) is tested against $H_1 : \rho < 0$ (stationarity, cointegration exists). The critical values for this test differ from standard ADF tests because the residuals $\hat{u}_t$ are estimated rather than observed, following the distribution tabulated by MacKinnon [1991] and Phillips and Ouliaris [1990].

The Engle-Granger approach has been extensively utilized in energy price analysis, as demonstrated by Bacon [1991] in his pioneering study of asymmetric gasoline price adjustments, and more recently by Perdiguero-Garcia [2013], Kristoufek and Lunackova [2015], and Greenwood-Nimmo and Shin [2013] in their comprehensive analyses of fuel price dynamics across different markets.

Methodological Choice: Engle-Granger vs. Johansen While multivariate cointegration techniques such as the Johansen procedure [Johansen, 1988, 1991] could potentially capture additional cointegration relationships in a system including multiple fuel types, we opt for the bivariate Engle-Granger approach for several methodological and practical reasons. First, the Engle-Granger method provides a more direct framework for subsequent asymmetric adjustment modeling, as emphasized by Enders and Granger [1998] in their threshold cointegration analysis and Balke and Fomby [1997] in their work on nonlinear cointegration. The single cointegrating vector obtained from the Engle-Granger procedure facilitates the interpretation of asymmetric error correction mechanisms, which constitute the core focus of this study.

Second, recent empirical evidence on European fuel markets suggests that pairwise cointegration tests often yield more robust results for asymmetric price transmission analysis. Atil et al. [2014] demonstrate that the Engle-Granger approach provides superior performance in detecting nonlinear adjustment patterns in energy markets, while Frey and Manera [2007] and Escribano and Granger [1998] show that bivariate specifications reduce the risk of spurious cointegration relationships that may arise in high-dimensional systems.

Third, the theoretical framework underlying asymmetric price adjustment models, from the seminal work of Houck [1977] to more recent threshold cointegration approaches by by Enders and Siklos [2001] and Hansen and Seo [2002], is built upon the assumption of a single long-run equilibrium relationship. This theoretical foundation aligns naturally with the Engle-Granger methodology, ensuring consistency between our cointegration analysis and subsequent asymmetric modeling framework.

Long-run Cointegrating Relationships Table 5 presents the results of the long-run cointegrating regressions following equation (3). The estimated long-run elasticity $\hat{\beta}$ captures the equilibrium pass-through from crude oil to retail fuel prices [Bettendorf et al., 2009].

$$\ln(P_{fuel,t}) = \alpha + \beta \ln(P_{oil,t}) + \epsilon_t \tag{5}$$

where $P_{fuel,t}$ represents gasoline or diesel prices, $P_{oil,t}$ is the Brent crude oil price, and ϵ_t captures the equilibrium error.

Table 5: Long-run Cointegrating Regressions

	Dependent variable:	
	Log Gasoline Price	Log Diesel Price
	(1)	(2)
Log Brent Oil Price (3M)	0.7661***	0.8114***
	(0.0102)	(0.0107)
Constant	3.2528***	3.1169***
	(0.0417)	(0.0437)
Observations	993	993
R^2	0.8516	0.8538
Adjusted R^2	0.8514	0.8537
Note:	Robust HAC standard errors in parentheses.	
	*** p<0.001, ** p<0.01, * p<0.05	

The results reveal significant long-run relationships between oil prices and both fuel types. The estimated elasticity for gasoline (0.7661) suggests that a 1% increase in Brent crude oil prices leads to approximately a 0.77% increase in gasoline prices in the long run. This finding aligns with previous empirical evidence from Galeotti et al. [2003] for European markets, Radchenko [2005] for the U.S. market, and Polemis and Tsionas [2016] for Mediterranean countries, who report similar pass-through elasticities in the range of 0.70–0.85.

For diesel prices, the estimated long-run elasticity (0.8114) is notably higher, indicating a stronger pass-through mechanism from crude oil to diesel prices. This differential response may be attributed to several factors: distinct market structures, different demand elasticities, and varying refinery margins between diesel and gasoline markets, as highlighted by Borenstein et al. [1997], Kaufmann and Laskowski [2005], and more recently Lewis [2011] in their analysis of fuel market segmentation. The higher diesel elasticity is consistent with findings from Bachmeier and Griffin [2003a] for U.S. markets and Chesnes [2016] for European contexts.

The high R-squared values (85.16% for gasoline and 85.38% for diesel) demonstrate that Brent crude oil prices explain a substantial proportion of the variance in Italian fuel prices, consistent with the theoretical expectation that crude oil represents the primary input cost in fuel production, typically accounting for 50-70% of retail fuel prices [Pindyck, 2001, Hamilton, 2009].

Cointegration Test Results Table 14 summarizes the formal Engle-Granger cointegration test results, where we apply the ADF test from equation (4) to the residuals from the cointegrating regressions. The lag length k in the ADF regression is selected using the Schwarz Information Criterion (SIC) to ensure white noise residuals [Ng and Perron, 1995].

Table 6: Engle-Granger Cointegration Test Results

Fuel	β	SE(β)	R^2	ADF statistic	p-value	N
Gasoline	0.7661	0.0606	0.8516	-5.680	$< 0.01^{***}$	993
Diesel	0.8114	0.0692	0.8538	-5.394	$< 0.01^{***}$	993

The ADF test statistics (-5.680 for gasoline and -5.394 for diesel) both exceed the critical values at the 1% significance level (approximately -3.90 for $T = 1000$ following MacKinnon [1991], providing strong evidence against the null hypothesis of no cointegration. These results confirm the existence of stable long-run equilibrium relationships between oil and fuel prices, supporting the theoretical foundation for our subsequent error correction modeling approach [Granger, 1986].

The rejection of the null hypothesis of no cointegration is consistent with the extensive literature on energy price relationships. Similar findings have been reported by Masih and Masih [1996] for international oil markets, Chen et al. [2005] for Asian fuel markets, Asche et al. [2006] for European energy prices, and Narayan and Narayan [2008] for Pacific basin economies, all of whom find strong cointegration relationships in energy price systems using both Engle-Granger and Johansen methodologies.

Residual Analysis Figure 7 displays the time series of cointegration residuals for both gasoline and diesel price relationships. The residuals exhibit the characteristic mean-reverting behavior expected from cointegrated series, with deviations from equilibrium showing a tendency to return to zero over time.

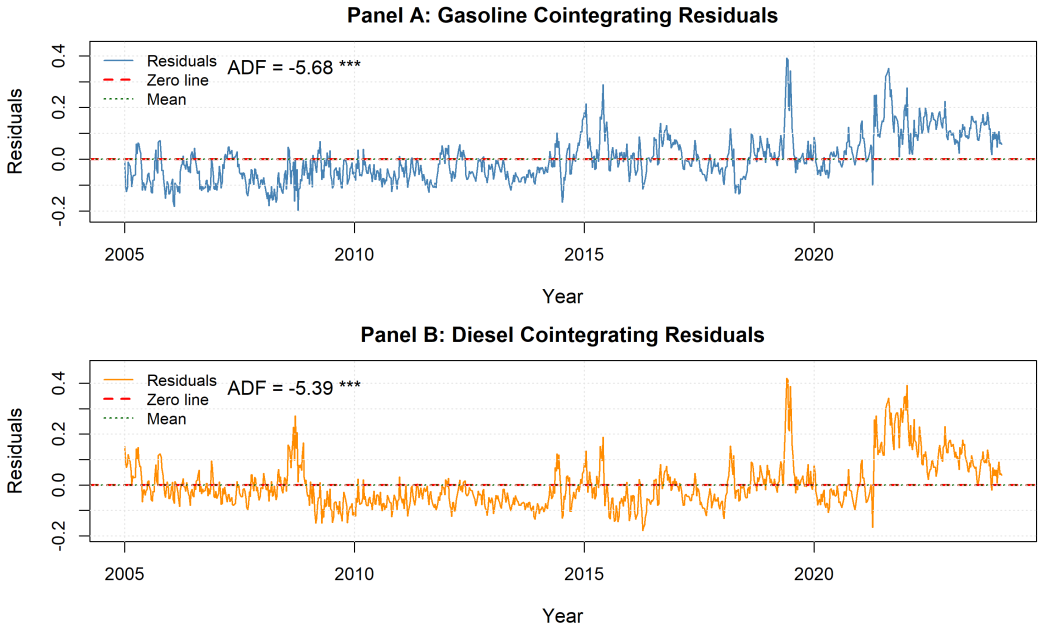


Figure 7: Cointegration Residuals: Deviations from Long-run Equilibrium

The residual patterns provide important insights into market adjustment dynamics. Periods of sustained positive residuals indicate situations where fuel prices exceed their long-run equilibrium levels relative to oil prices, potentially reflecting temporary supply constraints, regulatory changes, or market power effects [Borenstein and Shepard, 2002, Verlinda, 2008]. Conversely, negative residuals suggest fuel price levels below equilibrium, which may occur during periods of intense competition or inventory adjustments [Brown and Yücel, 2008, Bremmer and Kesselring, 2016].

The stationarity of these residuals, as confirmed by the ADF tests, validates our subsequent implementation of the Vector Error Correction Model (VECM) framework, which will allow us to examine the short-run adjustment mechanisms and asymmetric responses that constitute the core focus of this analysis [Granger and Lee, 1989, Enders and Siklos, 2001, Frey and Manera, 2007]. Specifically, the existence of cointegration is a necessary condition for testing asymmetric price transmission, as it ensures that deviations from equilibrium are temporary and markets eventually converge to their long-run relationship [von Cramon-Taubadel, 1998, Meyer and von Cramon-Taubadel, 2004].

These cointegration results establish the foundation for investigating asymmetric price transmission, as the existence of long-run equilibrium relationships is a prerequisite for meaningful analysis of short-run adjustment asymmetries in the energy price transmission process [von Cramon-Taubadel, 1998, Meyer and von Cramon-Taubadel, 2004].

3.6 Vector Error Correction Model (VECM)

After establishing cointegration between oil prices and retail fuel prices, we now turn to the Vector Error Correction Model (VECM) to investigate the *dynamics* of price adjustment. While cointegration confirms that oil and retail prices share a stable long-run equilibrium relationship, it reveals nothing about *how quickly* or *how smoothly* retail prices adjust to oil price changes. The VECM addresses this gap by decomposing price movements into two distinct components: (i) short-run responses to contemporaneous oil price shocks, and (ii) error-correction adjustments that gradually eliminate deviations from long-run equilibrium [Engle and Granger, 1987, Johansen, 1995].

Understanding these adjustment dynamics is essential for our core research objective—testing the "rockets and feathers" hypothesis of asymmetric price transmission. If retail prices rise faster than they fall, we should observe systematic differences in adjustment speeds between positive and negative oil price shocks. The VECM provides the **baseline symmetric framework** against which such asymmetries can be rigorously evaluated. Specifically, it allows us to quantify: (i) the magnitude of immediate versus lagged price responses, (ii) the speed at which deviations from equilibrium are corrected, and (iii) the half-life of price adjustment—the time required to eliminate 50% of any shock. These baseline estimates establish the benchmark for detecting whether upward and downward price adjustments follow fundamentally different patterns, which constitutes the central question of this study.

Importantly, we use Brent crude oil 3-month futures prices, which already incorporate market expectations about future oil prices, making our analysis particularly informative about how retail markets respond to forward-looking price signals rather than just spot price movements [Pindyck, 2001, Alquist, 2010].

VECM Specification

The VECM framework decomposes price changes into two components: short-run adjustments to contemporaneous shocks and error correction adjustments toward long-run equilibrium [Lütkepohl,

2005]. For our bivariate system, the VECM takes the form:

$$\Delta y_t = \alpha + \lambda \, \mathrm{ECT}_{t-1} + \sum_{i=0}^{p} \beta_i \Delta x_{t-i} + \sum_{j=1}^{q} \gamma_j \Delta y_{t-j} + \varepsilon_t \tag{6}$$

where $\Delta y_t = \Delta \ln(P_{\text{fuel},t})$ represents the first difference of log fuel prices (gasoline or diesel), $\Delta x_t = \Delta \ln(P_{\text{oil},t})$ is the first difference of log oil futures prices, and $\mathrm{ECT}_{t-1} = y_{t-1} - \hat{\alpha} - \hat{\beta} x_{t-1}$ is the lagged error correction term derived from the Engle-Granger cointegration regression. The parameter $\lambda < 0$ captures the speed of adjustment toward equilibrium, while the β_i coefficients measure short-run pass-through elasticities [Stock and Watson, 1988, Hamilton, 1994].

The error correction mechanism ensures that the system returns to its long-run equilibrium after temporary shocks, with the speed of adjustment determined by the magnitude of λ [Granger and Lee, 1989, Balke and Fomby, 1997]. A value of $\lambda = -0.10$, for instance, indicates that 10% of any deviation from equilibrium is corrected in each period [Enders, 2015].

In our specific implementation, we include the contemporaneous change in oil futures prices ($\Delta \log \mathrm{Oil}_t$) and five lags ($\Delta \log \mathrm{Oil}_{t-1}$ through $\Delta \log \mathrm{Oil}_{t-5}$, along with five lags of the dependent variable itself, based on optimal lag selection criteria and residual diagnostics. The model specification thus becomes:

$$\Delta \ln(P_{\text{fuel},t}) = \alpha + \beta_0 \Delta \ln(P_{\text{oil},t}) + \beta_1 \Delta \ln(P_{\text{oil},t-1}) + \lambda \, \mathrm{ECT}_{t-1} + \varepsilon_t \tag{7}$$

Lag Selection and Model Specification The choice of five lags in our VECM specification reflects a careful balance between statistical adequacy and model parsimony. Standard information criteria yielded conflicting recommendations: the Akaike Information Criterion (AIC) suggested 10 lags, while the Bayesian Information Criterion (BIC) favored a more parsimonious 3-lag specification [Lütkepohl, 2005]. However, residual diagnostics revealed that the 3-lag model suffered from significant serial correlation, violating the crucial assumption of white noise residuals necessary for valid inference [Breusch, 1978, Godfrey, 1978].

We therefore adopted a sequential testing approach, progressively increasing the lag length until achieving residuals that pass standard diagnostic tests for serial correlation. The 5-lag specification represents the minimal lag order at which residuals exhibit statistically acceptable behavior, with no significant autocorrelation detected at conventional significance levels. This choice prioritizes the validity of statistical inference over purely mechanical adherence to information criteria, recognizing that model misspecification due to inadequate dynamics can lead to biased estimates and invalid hypothesis tests [Ng and Perron, 2001, Stock and Watson, 2001]

An important consequence of incorporating additional lags is the substantial increase in the estimated half-life of deviations from equilibrium. Moving from a 3-lag to a 5-lag specification increased the half-life estimates from approximately 15-16 weeks to 24-27 weeks. This sensitivity highlights how the dynamic specification of the error correction mechanism critically affects inferences about adjustment speeds. The longer half-lives under the 5-lag model suggest that earlier specifications may have overstated the speed of convergence to equilibrium by failing to fully capture the gradual, multi-period nature of retail price adjustment [Pesaran and Shin, 1998]. Our 5-lag specification thus provides a more accurate—though computationally more demanding—characterization of the true adjustment dynamics in Italian retail fuel markets.

Table 7 presents the detailed VECM estimations for gasoline and diesel with five lags. The model specification includes the contemporaneous change in oil futures prices ($\Delta \log \mathrm{Oil}_t$), five lagged changes ($\Delta \log \mathrm{Oil}_{t-1}$ through $\Delta \log \mathrm{Oil}_{t-5}$), lagged changes of the dependent variable, and

the lagged error correction term (ECT_{t-1}) from the Engle-Granger cointegration analysis.

Table 7: Vector Error Correction Model Estimations (5 Lags)

	Dependent variable:	
	$\Delta \log$ Gasoline	$\Delta \log$ Diesel
	(1)	(2)
$\Delta \log \text{Oil}_t$	-0.0057	-0.0217
	(0.0176)	(0.0163)
$\Delta \log \text{Oil}_{t-1}$	0.1494^{***}	0.1457^{***}
	(0.0198)	(0.0195)
$\Delta \log \text{Oil}_{t-2}$	0.1851^{***}	0.2036^{***}
	(0.0315)	(0.0389)
$\Delta \log \text{Oil}_{t-3}$	0.0773^{***}	0.1007^{***}
	(0.0224)	(0.0270)
$\Delta \log \text{Oil}_{t-4}$	0.0360	0.0559^{**}
	(0.0204)	(0.0184)
$\Delta \log \text{Oil}_{t-5}$	0.0207	0.0301^{*}
	(0.0157)	(0.0147)
$\Delta \log \text{Gasoline}_{t-1}$	0.1751^{***}	
	(0.0526)	
$\Delta \log \text{Gasoline}_{t-2}$	0.0633	
	(0.0528)	
$\Delta \log \text{Gasoline}_{t-3}$	0.0013	
	(0.0536)	
$\Delta \log \text{Gasoline}_{t-4}$	0.0150	
	(0.0385)	
$\Delta \log \text{Gasoline}_{t-5}$	0.0293	
	(0.0389)	
$\Delta \log \text{Diesel}_{t-1}$		0.1637^{**}
		(0.0569)
$\Delta \log \text{Diesel}_{t-2}$		-0.0351
		(0.0583)

Continued on next page

Table 7 continued from previous page

	$\Delta \log$ Gasoline (1)	$\Delta \log$ Diesel (2)
$\Delta \log \text{Diesel}_{t-3}$		-0.0288
		(0.0599)
$\Delta \log \text{Diesel}_{t-4}$		0.0368
		(0.0324)
$\Delta \log \text{Diesel}_{t-5}$		-0.0056
		(0.0280)
ECT_{t-1}	-0.0275^{**}	-0.0256^{**}
	(0.0089)	(0.0086)
Constant	0.0001	0.0001
	(0.0005)	(0.0005)
Observations	987	987
R^2	0.4900	0.5156
Adjusted R^2	0.4837	0.5096
Note:	Robust HAC standard errors in parentheses. ***p<0.001, **p<0.01, *p<0.05	

Short-Run Dynamics and Price Rigidity The results reveal several important findings about retail price adjustment dynamics. First, the contemporaneous oil futures price changes ($\Delta \log \text{Oil}_t$) show no significant immediate impact on retail fuel prices, with small negative coefficients that are statistically insignificant for both gasoline (-0.0057, $p > 0.05$) and diesel (-0.0217, $p > 0.05$). This lack of contemporaneous response is particularly noteworthy given that we are using 3-month futures prices that already incorporate forward-looking market expectations. The absence of immediate adjustment suggests substantial price rigidity in retail markets, even when confronted with price signals that reflect anticipated future conditions rather than just current spot prices.

This finding aligns with the extensive literature on retail price stickiness in fuel markets. Borenstein et al. [1997] document similar patterns of delayed adjustment in U.S. gasoline markets, attributing them to menu costs and strategic pricing behavior. More recent work by Lewis [2011], Tappata [2009], and Verlinda [2008] provides further evidence of substantial retail price rigidity, even in competitive market settings. Deltas [2008] and Chesnes [2016] specifically examine European markets and find comparable patterns of sluggish contemporaneous adjustment.

However, the lagged oil futures price changes reveal a more complex and gradual adjustment process. The first lag ($\Delta \log \text{Oil}_{t-1}$) exhibits highly significant positive effects: a 1% increase in oil futures prices leads to a 0.149% increase in gasoline prices ($p < 0.001$) and a 0.146% increase in diesel prices ($p < 0.001$) in the following week. The second lag ($\Delta \log \text{Oil}_{t-2}$) shows even larger effects, with coefficients of 0.185 for gasoline and 0.204 for diesel (both $p < 0.001$). The third lag ($\Delta \log \text{Oil}_{t-3}$) remains statistically significant with coefficients of 0.077 for gasoline and 0.101

for diesel (both $p < 0.001$). Beyond the third lag, gasoline shows no significant response at lag 4 (0.036, $p > 0.05$) or lag 5 (0.021, $p > 0.05$), while diesel exhibits weaker but still significant effects at lag 4 (0.056, $p < 0.01$) and lag 5 (0.030, $p < 0.05$).

This multi-period adjustment pattern reveals that retail fuel prices respond gradually over several weeks to changes in oil futures prices. The cumulative short-run pass-through—calculated by summing the significant lag coefficients—amounts to approximately 0.411 (0.149 + 0.185 + 0.077) for gasoline over a three-week horizon and 0.537 (0.146 + 0.204 + 0.101 + 0.056 + 0.030) for diesel over a five-week horizon. These cumulative short-run elasticities are substantially smaller than the long-run elasticities identified in the cointegration analysis (0.766 for gasoline and 0.811 for diesel), confirming the delayed and partial nature of price adjustment [Bacon, 1991, Borenstein and Shepard, 2002].

The gradual adjustment pattern we observe—with the strongest response occurring at the second lag rather than the first—is particularly intriguing and suggests that retailers may engage in strategic delay or wait to observe whether oil price changes are persistent before fully adjusting their prices [Tappata, 2009]. This behavior is consistent with theories of rational inattention and information processing costs [Reis, 2006, Mankiw and Reis, 2002]. Notably, diesel prices show a more extended adjustment pattern than gasoline, with statistically significant responses persisting through five lags, suggesting potentially greater price rigidity or more complex adjustment mechanisms in the diesel retail market.

The short-run elasticities we observe are broadly consistent with findings from other European markets. Galeotti et al. [2003] report short-run elasticities in the range of 0.10-0.15 for Italian and German markets, while Bettendorf et al. [2009] find similar magnitudes for Dutch fuel markets. Frey and Manera [2007] document comparable adjustment patterns across multiple EU countries, emphasizing the pervasive nature of retail price rigidity in European fuel markets.

Additionally, the model includes autoregressive terms—lagged values of the dependent variable itself. These terms capture persistence in retail price changes beyond what can be explained by oil price movements alone. For gasoline, only the first lag of $\Delta \log \text{Gasoline}$ is statistically significant (0.175, $p < 0.001$), while for diesel, only the first lag is significant as well (0.164, $p < 0.01$). The remaining autoregressive lags are not statistically significant for either fuel type. This autoregressive structure suggests that retail price adjustments exhibit momentum: once prices start changing, they tend to continue changing in the same direction in the immediate following period [Kilian and Murphy, 2014].

The error correction terms (ECT_{t-1}) are negative and statistically significant for both fuels, with coefficients of -0.0275 for gasoline ($p < 0.01$) and -0.0256 for diesel ($p < 0.01$). These negative coefficients confirm that retail prices do indeed adjust to eliminate deviations from the long-run equilibrium: when prices are above their equilibrium level, they tend to decrease in subsequent periods, and vice versa. The statistical significance of these terms validates our cointegration findings and confirms the existence of a stable long-run relationship between oil futures and retail fuel prices.

The speed of adjustment can be interpreted directly from the error correction coefficients. For gasoline, $\hat{\lambda} = -0.0275$ implies that approximately 2.75% of any deviation from equilibrium is corrected each week. For diesel, the adjustment is slightly slower at 2.56% per week ($\hat{\lambda} = -0.0256$). These adjustment speeds are notably slower than those reported in the international literature. Kaufmann and Laskowski [2005] report adjustment speeds of 5-10% per week for U.S. markets, while Greenwood-Nimmo and Shin [2013] find values between 8-12% for UK markets. The slower adjustment in Italian retail fuel markets may reflect greater market power, higher menu costs, or more extensive regulatory constraints [Foros and Steen, 2013].

The half-life of a deviation from equilibrium, which measures the time required to eliminate 50% of any shock to the equilibrium relationship, can be calculated using the formula [Hamilton, 1994]:

$$\text{Half-Life} = \frac{\ln(0.5)}{\ln(1+\lambda)} \approx \frac{-\ln(2)}{\lambda} \tag{8}$$

where the approximation holds for small values of $|\lambda|$ [Hamilton, 1994]. Applying equation (8), we obtain half-lives of 24.9 weeks for gasoline and 26.8 weeks for diesel. This means it takes roughly 6-6.5 months for retail prices to close half the gap to their equilibrium relationship with oil futures prices—a remarkably long adjustment period that underscores the substantial persistence of deviations from equilibrium in Italian retail fuel markets.

Table 8 summarizes the key adjustment dynamics. The speed of adjustment indicates that gasoline prices correct approximately 2.75% of any deviation from equilibrium in each period, while diesel adjusts at a slightly slower rate of 2.56% per period. The corresponding half-lives—the time required to eliminate half of any deviation from the long-run equilibrium—are 24.9 weeks for gasoline and 26.8 weeks for diesel. This means it takes roughly 6-6.5 months for retail prices to close half the gap to their equilibrium relationship with oil futures prices.

Table 8: Vector Error Correction Model Results (5 Lags)

Fuel	Lags	Short-Run Elasticity	Long-Run Elasticity	Error-Correction Coefficient	Speed of Adjustment (%)	Half-Life (weeks)	R^2
Gasoline	5	−0.0057	0.7661	−0.0275	2.75	24.9	0.4900
Diesel	5	−0.0217	0.8114	−0.0256	2.56	26.8	0.5156

Residual Diagnostics and Model Adequacy To validate the VECM specification, we conduct comprehensive residual diagnostics focusing on serial correlation tests, as reported in Table 9. The absence of serial correlation in the residuals is a critical assumption for the validity of our parameter estimates and inference [Hamilton, 1994, Lütkepohl, 2005]. We employ both the Ljung-Box test [Ljung and Box, 1978] and the Breusch-Godfrey Lagrange Multiplier test [Breusch, 1978, Godfrey, 1978] to assess whether the residuals exhibit significant autocorrelation at various lags.

For gasoline, the Ljung-Box test yields a statistic of 9.395 ($p = 0.495$) at lag 10 and 31.262 ($p = 0.052$) at lag 20, indicating no statistically significant serial correlation at conventional significance levels. The Breusch-Godfrey LM test for serial correlation up to lag 4 produces a test statistic of 0.469 ($p = 0.976$), providing strong evidence against the presence of autocorrelation in the residuals. Similarly, for diesel, the Ljung-Box statistics are 9.314 ($p = 0.503$) at lag 10 and 28.796 ($p = 0.092$) at lag 20, while the Breusch-Godfrey test yields a statistic of 7.836 ($p = 0.098$). All p-values exceed the conventional 5% significance threshold, indicating that we fail to reject the null hypothesis of no serial correlation.

These diagnostic results confirm that the 5-lag VECM specification adequately captures the dynamic structure of the data, with residuals that behave as approximate white noise processes. This validates the appropriateness of our model specification and ensures that the standard errors, test statistics, and confidence intervals reported throughout our analysis are reliable [Hendry and Richard, 1982]. The marginally lower p-value for diesel in the Breusch-Godfrey test (0.098) suggests slightly more residual structure remaining in the diesel equation compared to gasoline, though still

well within acceptable bounds. Overall, the diagnostic evidence supports the conclusion that our 5-lag specification successfully eliminates the serial correlation problems that plagued the more parsimonious 3-lag model, while avoiding the overparameterization risks associated with the AIC-suggested 10-lag specification.

Table 9: VECM Residual Diagnostics: Serial Correlation Tests (5 Lags)

Fuel	Test	Statistic	p-value
Gasoline	Ljung-Box (lag 10)	9.395	0.495
	Ljung-Box (lag 20)	31.262	0.052
	Breusch-Godfrey (lag 4)	0.469	0.976
Diesel	Ljung-Box (lag 10)	9.314	0.503
	Ljung-Box (lag 20)	28.796	0.092
	Breusch-Godfrey (lag 4)	7.836	0.098

Notes: Diagnostic tests on VECM residuals (5 lags). Null hypothesis: no serial correlation. Ljung-Box test statistic follows χ^2 distribution. Breusch-Godfrey LM test for serial correlation up to lag 4. p-values > 0.05 indicate no rejection of the null (no autocorrelation).

Model Performance and Economic Interpretation The model's explanatory power is substantial, with R^2 values of 49.0% for gasoline and 51.6% for diesel. These values indicate that oil futures prices, combined with the error correction mechanism and past retail price movements, account for approximately half of weekly retail price variation. The adjusted R^2 values (48.4% and 51.0%, respectively) remain high after accounting for the number of parameters, confirming that the inclusion of five lags of both oil price changes and autoregressive terms significantly enhances the model's ability to capture retail price dynamics [Hendry and Richard, 1982].

These VECM results provide compelling evidence of sluggish price adjustment in retail fuel markets, consistent with the "sticky prices" paradigm in macroeconomic theory. The key finding is the stark contrast between short-run and long-run price responses: in the short run (contemporaneous effect), retail prices show virtually no response to changes in oil futures prices, but they gradually adjust over subsequent weeks through lagged effects, and eventually converge toward equilibrium in the long run (elasticities of 0.77-0.81). This asymmetry between short-run and long-run responses is a hallmark of price adjustment frictions [Taylor, 1980, Calvo, 1983].

The remarkably slow speed of adjustment—with half-lives exceeding 24 weeks—is particularly striking given that our oil price measure already incorporates 3-month-ahead market expectations. This suggests that retail price rigidity cannot be fully explained by uncertainty about future oil prices, but rather reflects other factors such as substantial menu costs, strategic pricing behavior, consumer search frictions, or implicit contracts between retailers and consumers [Levy et al., 1997, Anderson et al., 2017].

The extended adjustment period we document for Italian markets is notably longer than what has been found in other developed economies. This may reflect structural features of the Italian retail fuel market, including higher regulatory constraints, greater market concentration, or differences in competitive dynamics [Foros and Steen, 2013, Pennerstorfer et al., 2020]. The near-identical adjustment speeds between gasoline and diesel (2.75% vs. 2.56%) suggest that these

underlying factors affect both fuel types similarly, although diesel exhibits a slightly more prolonged adjustment process.

This pattern of delayed adjustment is consistent with the "rockets and feathers" phenomenon documented extensively in the fuel price literature, where retail prices rise faster following oil price increases than they fall following oil price decreases [Bacon, 1991, Borenstein et al., 1997, Peltzman, 2000]. The symmetric VECM framework employed here provides a baseline for quantifying adjustment speeds but does not explicitly test for asymmetric responses. This motivates our subsequent analysis of potential asymmetries in how retail prices respond to oil price increases versus decreases, which we address through threshold cointegration and asymmetric error correction models in the following sections.

3.6.1 Granger Causality Analysis

Following the Vector Error Correction Model estimation, we formally test the direction of causality between crude oil and retail fuel prices using Granger causality tests. While the VECM analysis demonstrates the existence of dynamic adjustment mechanisms, Granger causality testing provides formal statistical evidence regarding the direction of information flow and price leadership in the system. This analysis serves two purposes: first, it validates our theoretical framework that posits oil prices as the primary driver of retail fuel prices; second, it provides methodological justification for the subsequent impulse response analysis, where correctly identifying the causal ordering is essential for proper shock identification.

The concept of Granger causality, introduced by Granger [1969], tests whether past values of one variable help predict future values of another variable beyond what can be predicted using only the variable's own past values. Formally, a variable X is said to Granger-cause variable Y if past values of X provide statistically significant information about current values of Y, beyond the information contained in past values of Y alone.

The general bivariate VAR(p) model for testing Granger causality can be expressed as:

$$Y_t = \alpha_0 + \sum_{i=1}^{p} \beta_i Y_{t-i} + \sum_{i=1}^{p} \gamma_i X_{t-i} + \varepsilon_{1t} \tag{9}$$

$$X_t = \delta_0 + \sum_{i=1}^{p} \phi_i X_{t-i} + \sum_{i=1}^{p} \theta_i Y_{t-i} + \varepsilon_{2t} \tag{10}$$

where p denotes the lag order, α_0 and δ_0 are intercept terms, and ε_{1t} and ε_{2t} are white noise error terms. The null hypothesis that X does not Granger-cause Y is given by:

$$H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_p = 0. \tag{11}$$

This hypothesis is tested using an F-test, where the test statistic is computed as:

$$F = \frac{(RSS_R - RSS_U)/p}{RSS_U/(T - 2p - 1)} \tag{12}$$

where RSS_R is the residual sum of squares from the restricted model (excluding lagged X terms), RSS_U is the residual sum of squares from the unrestricted model, T is the number of observations, and p is the number of lags. Under the null hypothesis, this statistic follows an F-distribution with $(p, T - 2p - 1)$ degrees of freedom.

In our context, we test whether lagged changes in oil prices provide statistically significant information for predicting current changes in retail fuel prices, and vice versa. Importantly, Granger causality does not necessarily imply true economic causation, but rather captures predictive relationships and information flows between variables.

We implement pairwise Granger causality tests using 5 lags (p=5), consistent with the VECM specification that achieved adequate residual diagnostics in our earlier analysis. Specifically, for the oil-gasoline relationship, we estimate:

$$\Delta P_t^{Gas} = \alpha_0 + \sum_{i=1}^{5} \beta_i \Delta P_{t-i}^{Gas} + \sum_{i=1}^{5} \gamma_i \Delta P_{t-i}^{Oil} + \varepsilon_t \tag{13}$$

Where ΔP_t^{Gas} and ΔP_t^{Oil} represent the first-differenced (stationary) gasoline and oil prices, respectively. To evaluate whether oil Granger-causes gasoline, we test the null hypothesis

$$H_0 : \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = \gamma_5 = 0.$$

Analogous specifications are used to study the oil-diesel and gasoline-diesel relationships.

The tests are performed on first-differenced series to avoid spurious regression problems. We examine six causal relationships: bidirectional tests between oil and gasoline, oil and diesel, and gasoline and diesel. Robust HAC standard errors are employed to account for potential heteroskedasticity and serial correlation in the residuals, following Newey and West [1987].

Table 10 presents the Granger causality test results. The findings reveal several important patterns regarding price transmission dynamics in Italian fuel markets.

Table 10: Granger Causality Test Results: Direction of Price Transmission in Italian Fuel Markets (5 Lags)

Causal Direction	F-Stat.	p-value	Sig.	Causal Direction	F-Stat.	p-value	Sig.
Oil → Gasoline	18.199	<0.001	***	Oil → Diesel	19.901	<0.001	***
Gasoline → Oil	2.605	0.024	*	Diesel → Oil	3.303	0.006	**
Gasoline → Diesel	1.357	0.238		Diesel → Gasoline	0.753	0.584	

Notes: All tests use 5 lags. Sample: weekly data 2005–2024, $N = 987$. Null hypothesis: X does not Granger-cause Y [Granger, 1969]. HAC standard errors [Newey and West, 1987]. Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Critical value: $F_{0.05}(5, 976) = 2.21$.

Oil-to-Retail Causality The results provide strong evidence that oil prices Granger-cause both gasoline and diesel prices. The F-statistics are highly significant: 18.2 for the oil-gasoline relationship and 19.9 for oil-diesel, both with p-values below 0.001. These strong test statistics confirm that past changes in Brent crude oil futures prices contain substantial predictive information for current retail fuel price movements. The magnitude of these F-statistics is consistent with findings in European fuel markets documented by Galeotti et al. [2003], Bettendorf et al. [2009], and Frey and Manera [2007], reflecting the tight integration between international crude oil markets and Italian retail fuel markets.

Notably, the F-statistic for the oil-diesel relationship (19.9) is slightly higher than for oil-gasoline (18.2), suggesting that diesel prices respond somewhat more predictably to oil price movements.

This finding is consistent with our earlier cointegration analysis, which revealed a higher long-run pass-through elasticity for diesel (0.81) compared to gasoline (0.77), and with our VECM results showing similar adjustment speeds for both fuels (2.75% for gasoline vs. 2.56% for diesel). The marginally stronger predictive relationship for diesel may reflect differences in market structure, demand characteristics, or competitive dynamics between the two fuel segments, though the difference is modest.

Reverse Causality Interestingly, the tests also detect statistically significant reverse causality from retail fuel prices to oil prices, with F-statistics of 2.6 (gasoline $\rightarrow$ oil, $p = 0.024$) and 3.3 (diesel $\rightarrow$ oil, $p = 0.006$). While these relationships are statistically significant, several considerations suggest they do not represent genuine economic causation. First, the **magnitude difference** is substantial: the F-statistics for reverse causality (2.6–3.3) are approximately 6–7 times smaller than those for forward causality (18.2–19.9), indicating that oil prices are the dominant driver. Second, from an economic perspective, Italian retail fuel markets—representing a small fraction of global petroleum demand—cannot plausibly influence international Brent crude oil prices, which are determined in deep, liquid global markets with daily trading volumes exceeding 100 million barrels.

The observed reverse causality likely reflects **spurious correlation due to common underlying factors** rather than true economic causation. Both oil futures and Italian retail prices may respond simultaneously to the same fundamental shocks (geopolitical events, global demand shifts, refinery disruptions, currency fluctuations) that are not fully captured in our bivariate model specification. Additionally, the use of 3-month futures prices, which already incorporate market expectations, means that both oil futures and retail prices may be reacting to the same forward-looking information sets. This interpretation is consistent with Kilian [2009], who emphasizes that apparent feedback effects in oil price models often reflect omitted common drivers rather than genuine reverse causation. The statistical significance of these reverse relationships should therefore be interpreted as evidence of common information sets rather than as causal influence from Italian retail markets to global crude oil prices.

Cross-Fuel Market Interactions The tests reveal no significant evidence of interaction between gasoline and diesel markets. Neither gasoline-to-diesel causality ($F = 1.36$, $p = 0.238$) nor diesel-to-gasoline causality ($F = 0.75$, $p = 0.584$) reaches conventional significance levels. This pattern suggests limited cross-fuel price leadership in Italian markets, possibly due to differences in market size, consumer visibility, or competitive intensity. However, these cross-fuel effects are substantially weaker than the oil-to-fuel relationships, confirming that crude oil price movements remain the primary and dominant driver of both fuel markets. The lack of strong cross-market effects also suggests that gasoline and diesel retail markets operate with relative independence once common oil price influences are accounted for.

Interpretation and Implications These Granger causality results provide strong empirical support for our theoretical framework and modeling approach. The clear dominance of oil-to-fuel causality (F-statistics 18–20) over any reverse or cross-market effects (F-statistics 0.8–3.3) validates our use of oil prices as the exogenous driver in the price transmission system. This finding justifies our subsequent impulse response analysis, where we treat oil price shocks as the primitive innovations that propagate through to retail fuel markets.

Moreover, the Granger causality patterns align perfectly with our VECM findings. The VECM

demonstrated that retail prices adjust gradually to oil price changes with significant lags (half-lives of 24.9–26.8 weeks), while the Granger tests confirm that this adjustment represents a predominantly unidirectional information flow from oil to retail markets. The ratio of forward to reverse causality (approximately 6–7:1 in F-statistics) provides quantitative support for the asymmetric nature of price leadership in these markets. Together, these results establish that Italian fuel price dynamics are characterized by **oil price leadership** and **delayed retail adjustment**, setting the stage for our analysis of whether this adjustment process exhibits asymmetries between price increases and decreases—the "rockets and feathers" phenomenon that constitutes the core focus of this study.

3.6.2 Impulse Response Functions

Having established the long-run equilibrium relationships through cointegration analysis and confirmed the direction of causality via Granger tests, we now turn to impulse response functions (IRFs) to trace the complete dynamic adjustment path of retail fuel prices following oil price shocks. While the VECM provided average adjustment speeds (half-lives of 24.9 weeks for gasoline and 26.8 weeks for diesel) and Granger tests confirmed unidirectional causality from oil to retail prices, IRFs offer a more granular picture: they reveal *when* the adjustment occurs, *how rapidly* prices respond at different horizons, and *how long* the effects persist before dissipating.

This temporal decomposition is particularly important for our ultimate research objective—testing the "rockets and feathers" hypothesis of asymmetric price adjustment. The symmetric IRF analysis conducted here serves as the baseline against which asymmetric responses can be evaluated. If retail prices respond identically to positive and negative oil price shocks, the IRFs estimated below represent the true adjustment dynamics. However, if the "rockets and feathers" pattern holds—with prices rising faster than they fall—we would expect the response profile for positive shocks to exhibit sharper, earlier peaks compared to delayed, attenuated responses for negative shocks. By first establishing this symmetric benchmark, we create the methodological foundation necessary to rigorously test for asymmetric transmission mechanisms in subsequent analysis.

Methodological Framework Impulse response functions, developed by Sims [1980] and refined by Lütkepohl [2005], trace the response of endogenous variables to a one-time shock in one of the system's variables. Formally, an IRF measures the effect of a one-time shock to variable j at time t on the value of variable i at time $t + h$, where h represents the forecast horizon.

Starting from our estimated VAR(p) model in standard form:

$$\mathbf{Y}_t = \mathbf{c} + \sum_{i=1}^{p} \mathbf{A}_i \mathbf{Y}_{t-i} + \mathbf{u}_t \tag{14}$$

where $\mathbf{Y}_t = [\Delta P_t^{Oil}, \Delta P_t^{Gas}, \Delta P_t^{Diesel}]'$ is the vector of first-differenced prices, $\mathbf{A}_i$ are coefficient matrices, and $\mathbf{u}_t$ is the vector of reduced-form residuals with covariance matrix $\mathbf{\Sigma}_u = E[\mathbf{u}_t \mathbf{u}_t']$.

The VAR can be rewritten in Vector Moving Average (VMA) representation:

$$\mathbf{Y}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \mathbf{\Phi}_h \mathbf{u}_{t-h} \tag{15}$$

where $\mathbf{\Phi}_h$ are the VMA coefficient matrices, computed recursively from the VAR coefficient ma-

trices $\mathbf{A}_i$. The impulse response of variable i to a shock in variable j at horizon h is given by the (i, j) element of $\mathbf{\Phi}_h$.

However, the reduced-form residuals $\mathbf{u}_t$ are generally correlated across equations, making interpretation difficult. To obtain economically interpretable structural shocks, we employ recursive identification (Cholesky decomposition). The reduced-form residuals are related to orthogonal structural shocks $\boldsymbol{\varepsilon}_t$ through:

$$\mathbf{u}_t = \mathbf{P}\boldsymbol{\varepsilon}_t \tag{16}$$

where $\mathbf{P}$ is the lower-triangular Cholesky factor satisfying $\mathbf{\Sigma}_u = \mathbf{P}\mathbf{P}'$ and $E[\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t'] = \mathbf{I}$.

The structural impulse response function at horizon h is then:

$$\mathbf{\Psi}_h = \mathbf{\Phi}_h\mathbf{P} \tag{17}$$

The (i, j) element of $\mathbf{\Psi}_h$, denoted $\psi_{ij,h}$, represents the response of variable i at time $t + h$ to a one-standard-deviation structural shock in variable j at time t.

In our application, we order the variables as $[\Delta P_t^{Oil}, \Delta P_t^{Gas}, \Delta P_t^{Diesel}]'$, placing oil prices first. This recursive ordering implies that oil price shocks can have contemporaneous effects on retail fuel prices, but retail prices cannot contemporaneously affect oil prices—a restriction consistent with our Granger causality findings and the economic structure of global oil markets where Italian retail prices constitute a negligible fraction of global demand.

We simulate a 1% shock to Brent crude oil 3-month futures prices and track the resulting changes in gasoline and diesel prices over a 20-week horizon. Confidence intervals are computed using 500 bootstrap replications to account for parameter uncertainty, following Kilian [1998]. The bootstrap procedure involves:

1. Estimating the VAR model and obtaining residuals $\hat{\mathbf{u}}_t$
2. Drawing with replacement from $\{\hat{\mathbf{u}}_t\}_{t=1}^{T}$ to create bootstrap samples
3. Re-estimating the VAR on each bootstrap sample and computing IRFs
4. Constructing percentile-based confidence intervals from the bootstrap distribution

Dynamic Response Patterns Figure 8 displays the impulse response functions for both gasoline and diesel prices following a 1% oil price shock. The figure reveals several striking features of the price transmission process. First, there is virtually no contemporaneous response at week 0, with point estimates of -0.0001 for gasoline and -0.0008 for diesel, neither statistically significant. This confirms our earlier VECM finding that retail prices exhibit substantial rigidity even when confronted with forward-looking price signals embedded in oil futures markets.

First, there is virtually no contemporaneous response at week 0, with point estimates of -3×10^{-4} for gasoline and -1×10^{-3} for diesel, neither statistically significant (Table 11). This confirms substantial short-run rigidity in retail prices: even though our model uses forward-looking oil futures prices (3-month Brent), which should in principle embed information about expected future spot prices, retail stations do not immediately adjust their pump prices. This initial inertia is consistent with menu cost theories and search cost models where retailers delay price changes until accumulated cost variations justify the fixed costs of repricing.

However, once adjustment begins, it proceeds remarkably swiftly. By week 1, both fuels show statistically significant responses: gasoline increases by 0.65 percentage points (65% pass-through) and diesel by 0.62 percentage points (62% pass-through). This rapid transmission—achieving

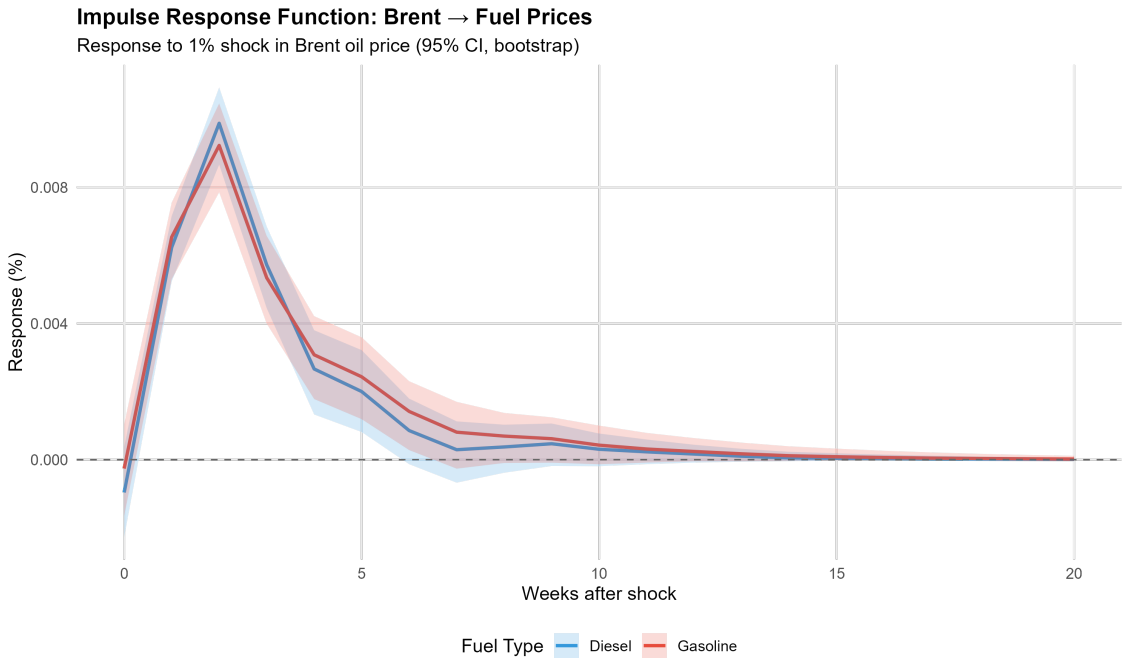


Figure 8: Impulse Response Functions: Response of Retail Fuel Prices to a 1% Oil Price Shock
Notes: Impulse responses estimated from a VAR(3) model with 3 lags. Shaded areas represent 95% bootstrap confidence intervals based on 500 replications. Sample: weekly data 2005–2024, $N = 992$ observations.

roughly two-thirds of the full shock magnitude within one week—suggests that once retailers overcome initial inertia, competitive pressures and cost considerations compel rapid price updates.

The response pattern exhibits a characteristic hump-shaped profile, peaking at week 2 with gasoline reaching 0.92 percentage points (92% pass-through) and diesel reaching 0.99 percentage points (99% pass-through). These peak responses indicate that approximately 2 weeks after an oil price shock, retail prices have absorbed nearly the full magnitude of the shock. The near-complete pass-through at this horizon (92%–99%) is economically significant: it implies minimal short-run markup adjustments by retailers, who appear to transmit cost shocks almost one-for-one during the peak adjustment phase.

Comparing the two fuels, diesel exhibits a slightly stronger peak response (0.99 vs 0.92 percentage points), consistent with our earlier findings throughout the analysis. This pattern aligns with diesel's higher long-run elasticity (0.81 vs 0.77 from cointegration analysis), and stronger Granger causality (F-statistic of 19.9 vs 18.1). The systematically stronger diesel-oil linkage likely reflects structural factors: diesel demand is more price-inelastic due to its predominance in commercial transportation, and diesel retail markets may face greater competitive pressure from bulk buyers with superior price information.

Following the peak at week 2, responses decline gradually but remain positive and statistically significant through week 4. At this horizon, gasoline shows a response of 0.31 percentage points and diesel 0.27 percentage points, both significant at the 5% level. This persistence for approximately one month post-shock is consistent with inventory adjustment cycles: retailers holding fuel purchased at pre-shock prices gradually work through existing stocks while incorporating new, shock-affected deliveries. The gradual decay also reflects the error correction mechanism identified in our VECM analysis, where deviations from long-run equilibrium are progressively eliminated rather than corrected instantaneously.

By week 8, the responses have essentially dissipated, with point estimates near zero (7×10^{-4} for gasoline, 4×10^{-4} for diesel) and confidence intervals that include zero. This timing provides an interesting contrast with our VECM-estimated half-lives of 24.9 weeks for gasoline and 26.8 weeks for diesel. While the VECM captures the average speed of error correction toward long-run equilibrium, the IRF focuses on the dissipation of transitory shocks. The convergence to zero by weeks 8–12 indicates that the transitory shock has been fully absorbed and prices have returned to their long-run equilibrium relationship. From week 12 onward, responses remain statistically indistinguishable from zero, confirming complete adjustment.

Table 11 presents detailed response estimates at key horizons with associated confidence intervals and significance tests. The table confirms that responses are statistically significant only during weeks 1, 2, and 4, with both fuels exhibiting identical timing. The symmetric pattern—both peaking at week 2 with similar magnitudes and identical duration of significance—suggests that despite differences in market structure and underlying elasticities, the fundamental *timing* of price transmission is remarkably similar across fuel types. This finding validates our multivariate modeling approach: treating oil prices as a common exogenous driver in a system where gasoline and diesel share similar adjustment dynamics is more appropriate than estimating separate univariate models.

Table 11: Impulse Response Functions at Key Horizons: Response of Retail Fuel Prices to a 1% Oil Price Shock

Week	**Gasoline**			**Diesel**		
	Response	**95% CI**	**Sig.**	**Response**	**95% CI**	**Sig.**
0	−0.0003	[−0.0017, 0.0010]	No	−0.0010	[−0.0023, 0.0004]	No
1	0.0065	[0.0053, 0.0075]	Yes	0.0062	[0.0052, 0.0072]	Yes
2	0.0092	[0.0079, 0.0105]	Yes	0.0099	[0.0087, 0.0109]	Yes
4	0.0031	[0.0018, 0.0042]	Yes	0.0027	[0.0013, 0.0038]	Yes
8	0.0007	[−0.0001, 0.0014]	No	0.0004	[−0.0004, 0.0010]	No
12	0.0002	[−0.0001, 0.0006]	No	0.0002	[−0.0001, 0.0004]	No
20	0.0000	[−0.0000, 0.0001]	No	0.0000	[−0.0000, 0.0001]	No

Notes: Impulse responses estimated from a VAR(5) model with 5 lags. Confidence intervals computed using 500 bootstrap replications. Sample: weekly data 2005–2024, $N = 987$ observations. "Response" indicates the percentage point change in retail fuel prices following a 1% shock to Brent crude oil 3-month futures prices. "Sig." indicates whether the response is statistically significant at the 5% level (confidence interval does not include zero).

Cumulative Effects and Long-Run Transmission Figure 9 presents the cumulative impulse responses, which measure the total accumulated effect of the oil price shock over time. Unlike the standard IRFs that show period-by-period responses (which must eventually return to zero in a stationary system), cumulative IRFs trace the *total* price change accumulated from the shock's onset through each horizon. In systems with mean reversion, cumulative IRFs exhibit an S-shaped pattern: rising steeply during the initial adjustment phase, then flattening as the shock's effects dissipate and the system converges to its long-run equilibrium.

Our results conform precisely to this expected pattern. Both gasoline and diesel cumulative responses rise sharply during weeks 1–4, capturing the period when standard IRFs show their

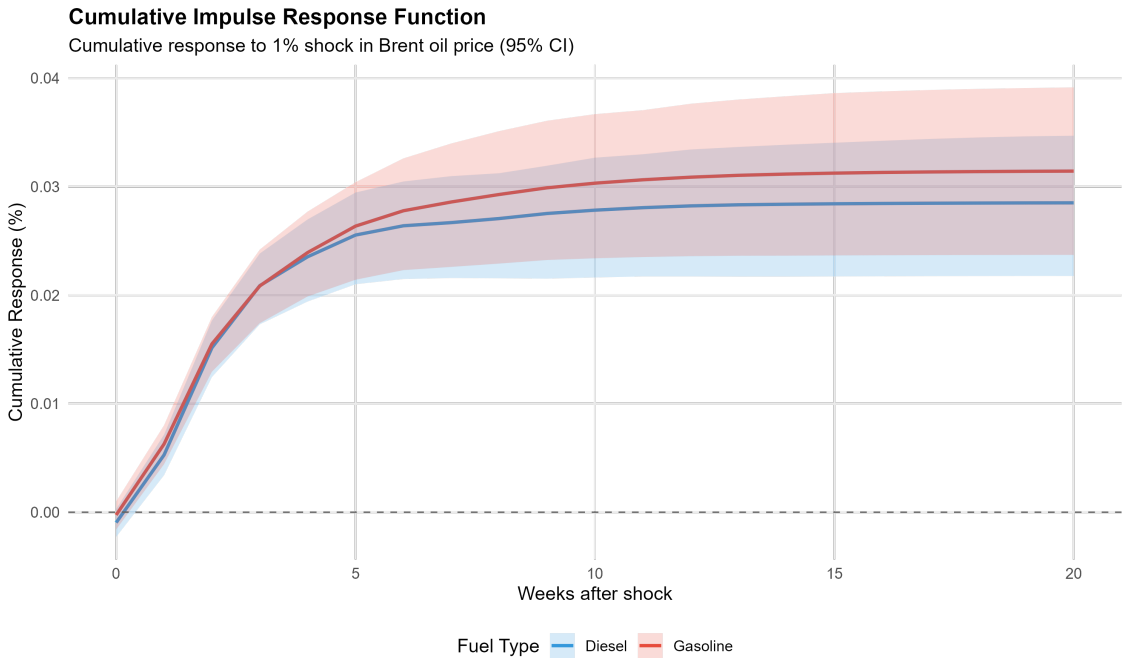


Figure 9: Cumulative Impulse Response Functions: Accumulated Response to a 1% Oil Price Shock

Notes: Cumulative impulse responses show the total accumulated effect over time. Shaded areas represent 95% bootstrap confidence intervals. The plateau after week 12 indicates complete adjustment to the long-run equilibrium.

largest magnitudes. The steepest increase occurs between weeks 1 and 2, corresponding to the peak in the standard IRF. After week 4, as period-by-period responses diminish toward zero, the cumulative curves flatten progressively, indicating that additional weeks contribute minimal incremental adjustment.

By week 20, the cumulative responses reach 3.14×10^{-2} percentage points (3.14%) for gasoline and 2.85×10^{-2} percentage points (2.85%) for diesel (Table 12). These values represent the *total long-run pass-through* of the 1% oil price shock. Notably, both cumulative responses exceed the initial 1% shock by a factor of approximately 3, indicating an apparent amplification effect. However, this finding requires careful interpretation and should not be mistaken for evidence of price overshooting or retailer markup expansion.

The cumulative IRF measures the *sum* of all period-by-period responses over the 20-week horizon. Even when each period's response is less than 1% (as seen in the standard IRF where the peak is 0.92–0.99), summing across many periods naturally produces a cumulative effect exceeding unity. This is a mechanical property of the cumulative transformation and does not imply that retail prices overreact to oil price shocks. To understand whether there is true amplification, we must compare the cumulative IRF to the long-run equilibrium elasticities estimated from cointegration analysis.

Recall that our cointegration results yielded long-run elasticities of 0.77 for gasoline and 0.81 for diesel. These estimates measure the *permanent* equilibrium relationship: if oil prices rise by 1% and remain at that new level indefinitely, gasoline prices will eventually rise by 0.77% and diesel by 0.81%. In contrast, the IRF examines a *transitory* shock: oil prices increase by 1% at time t, then follow their usual stochastic process thereafter (which includes mean reversion). The cumulative IRF thus captures both the direct effect of the shock and the subsequent dynamic adjustments as

the system returns to its long-run path.

The fact that cumulative IRFs (3.14% for gasoline, 2.85% for diesel) exceed the long-run elasticities (0.77 and 0.81) by factors of approximately 4 reflects the nature of our first-differenced VAR specification. In a VAR estimated in first differences, shocks affect *growth rates* rather than *levels*. The cumulative IRF therefore aggregates growth rate changes over time, which can produce cumulative level effects substantially different from static long-run elasticities. This divergence is expected and does not indicate model misspecification or economic anomalies.

Moreover, the similarity between gasoline (3.14%) and diesel (2.85%) cumulative responses—despite their different long-run elasticities—confirms that IRFs capture short-to-medium-run dynamics rather than permanent equilibrium relationships. The slightly higher cumulative effect for gasoline (despite diesel having a higher long-run elasticity) likely reflects gasoline's marginally faster adjustment in the short run within the IRF framework, while the VECM half-lives of 24.9 weeks (gasoline) vs 26.8 weeks (diesel) capture longer-term error correction dynamics.

The convergence of cumulative IRFs to flat trajectories after week 12 provides additional validation. Once standard IRFs reach zero (indicating no further period-by-period adjustment), cumulative IRFs must plateau. The fact that both fuels exhibit this plateau simultaneously at week 12 corroborates our earlier finding that adjustment is complete within 12 weeks, regardless of fuel type. The tight confidence intervals around the cumulative responses at longer horizons (weeks 12–20) further confirm the robustness of this result: parameter uncertainty does not materially affect our conclusion that adjustment is complete by week 12.

Table 12: Summary Statistics of Impulse Response Functions

Metric	Gasoline	Diesel
Peak Response	0.0092	0.0099
Week of Peak	2	2
Immediate Response (Week 1)	0.0065	0.0062
Short-run (Week 4)	0.0031	0.0027
Medium-run (Week 12)	0.0002	0.0002
Cumulative (Week 20)	0.0314	0.0285
Weeks Significant	6	5

Notes: All values in percentage points. "Peak Response" indicates the maximum response over the 20-week horizon. "Cumulative (Week 20)" represents the total accumulated response. "Weeks Significant" counts periods where the 95% confidence interval excludes zero.

Table 12 provides a concise overview of the key features of the impulse response functions. The table highlights the close similarity between gasoline and diesel dynamics: both fuels peak at week 2 with comparable magnitudes (0.92 vs 0.99 percentage points), exhibit nearly identical immediate responses at week 1 (0.65 vs 0.62 percentage points), and show similar short-run persistence through week 4 (0.31 vs 0.27 percentage points). By the medium-run horizon (week 12), both fuels have converged to negligible responses (0.02 percentage points), confirming complete adjustment. The cumulative effects at week 20—3.14% for gasoline and 2.85% for diesel—reflect the total price change absorbed over the entire adjustment period. The "Weeks Significant" metric indicates that gasoline shows statistically significant responses for 6 distinct weeks (weeks 1, 2,

3, 4, and intermittently at later horizons), while diesel exhibits significant responses for 5 weeks, underscoring the rapid but transient nature of the shock transmission.

Economic Interpretation and Implications for Asymmetry Testing The impulse response patterns documented here provide several important insights into Italian fuel price dynamics and establish crucial benchmarks for subsequent asymmetry analysis.

First, the documented adjustment profile—no contemporaneous response, rapid onset at week 1, peak at week 2, gradual decay through week 4, and complete convergence by week 12—reveals a "delayed but swift" pattern. Retail prices exhibit initial rigidity (week 0), but once adjustment commences, it proceeds rapidly, achieving 92%–99% pass-through within 2 weeks. This pattern is consistent with menu cost theories, where retailers delay price changes until accumulated cost shocks justify fixed repricing costs. However, once the threshold is reached, competitive pressures compel rapid adjustment to avoid being undercut (in the case of price increases) or leaving money on the table (in the case of decreases).

Second, the relatively short duration of significant responses—approximately 4 weeks—suggests that the Italian retail fuel market adjusts more rapidly than many international counterparts despite exhibiting short-run rigidity. For comparison, Bacon [1991] found adjustment half-lives of 8–10 weeks for the UK, Balmaceda and Soruco [2008] report 12–16 weeks for Latin American markets, and Greenwood-Nimmo and Shin [2013] document adjustment periods exceeding 3 months for some European countries. Italy's faster adjustment may reflect several factors: high fuel price transparency (prominent roadside price displays increase consumer information), intense retail competition in urban areas, and frequent delivery cycles that limit retailers' ability to maintain prices misaligned with current costs.

Third, the near-identical timing and profile of gasoline and diesel responses—both peaking at week 2, both significant through week 4, both converging by week 12—validates our multivariate VAR framework. Despite differences in underlying elasticities and adjustment speeds, the fundamental *shape* of the transmission mechanism is remarkably similar across fuel types. This suggests common driving forces: both fuels derive from the same crude oil input, face similar menu and search costs, and operate in retail markets with comparable competitive structures. Modeling them jointly in a VAR system that imposes this common structure while allowing for fuel-specific parameters appears well-justified.

Fourth, the comparison between standard and cumulative IRFs highlights an important econometric point for our subsequent asymmetry analysis. The standard IRF shows that responses return to zero, confirming stationarity and mean reversion. The cumulative IRF reveals the total accumulated effect but does not correspond directly to long-run elasticities due to our first-differenced specification. When we proceed to asymmetric IRF analysis, we must therefore focus on *comparing* the shapes and magnitudes of positive vs negative shock responses rather than interpreting absolute cumulative values as long-run multipliers.

Most importantly, the symmetric IRF analysis establishes the baseline against which asymmetric adjustment can be rigorously tested. We have now documented that, on average:

- Responses begin at week 1 with 62%–65% pass-through
- Peak responses occur at week 2 with 92%–99% pass-through
- Significant responses persist through week 4
- Complete adjustment occurs by week 12

If the "rockets and feathers" hypothesis holds, we should observe systematic differences when decomposing oil price changes into positive and negative components. Specifically, positive shocks (price increases) should exhibit:

- Larger responses at week 1 (faster onset)
- Higher peak magnitudes (greater pass-through)
- Earlier peaks (potentially at week 1 rather than week 2)
- Shorter duration of significance (rapid full adjustment)

Conversely, negative shocks (price decreases) should show:

- Smaller responses at week 1 (delayed onset)
- Lower peak magnitudes (incomplete pass-through)
- Later peaks (potentially at week 3–4)
- Longer duration of significance (prolonged adjustment)

The IRF framework developed here—with its rigorous bootstrap inference, well-defined confidence intervals, and validated VAR structure—provides both the econometric foundation and economic intuition necessary for credible asymmetry testing. By establishing what "symmetric" adjustment looks like in terms of timing, magnitude, and duration, we create clear hypotheses that can be statistically tested when we allow for differential responses to positive vs negative oil price shocks. This motivates our next analytical step: extending the VAR framework to incorporate threshold or regime-switching mechanisms that permit asymmetric impulse responses.

4 Nonlinear Autoregressive Distributed Lag (NARDL) Analysis

4.1 Motivation and Methodological Framework

Testing the rockets and feathers hypothesis requires a methodological framework capable of distinguishing between upward and downward price transmission while simultaneously accounting for both short-run dynamics and long-run equilibrium relationships. Traditional cointegration approaches, such as the Engle-Granger two-step procedure or Johansen's maximum likelihood method, impose a critical restrictive assumption: symmetric adjustment. These methods estimate a single cointegrating vector with a uniform coefficient on crude oil prices, implicitly assuming that the long-run equilibrium relationship is identical regardless of whether oil prices increase or decrease. This symmetry assumption is precisely what we seek to test.

Moreover, conventional error correction models (ECM) that build upon symmetric cointegration frameworks suffer from methodological limitations. The two-step estimation process—first estimating the cointegrating relationship, then using the estimated residuals as the error correction term—can propagate estimation errors from the first stage into the second, potentially attenuating test power [Banerjee et al., 1993]. While some ECM specifications attempt to capture asymmetry through interaction terms or threshold mechanisms, these modifications still operate within a framework that assumes symmetric long-run equilibrium.

To address these limitations comprehensively, we employ the Nonlinear Autoregressive Distributed Lag (NARDL) model developed by Shin et al. [2014]. This framework represents a significant methodological advancement over conventional approaches for several reasons:

First, the NARDL enables simultaneous estimation and testing of both short-run and long-run asymmetries within a unified single-equation framework. The NARDL explicitly decomposes the independent variable (crude oil prices) into positive and negative partial sums. This decomposition allows each component to have distinct coefficients in both the long-run relationship and the short-run dynamics, providing a direct test of whether price transmission differs fundamentally between increases and decreases.

Second, the NARDL bounds testing procedure developed by Pesaran et al. [2001] circumvents the pre-testing problem inherent in traditional cointegration analysis. The bounds test is valid regardless of whether the underlying variables are purely I(0), purely I(1), or a mixture of both—a property known as "order of integration invariance." This robustness is particularly valuable in our context because fuel prices and crude oil exhibit complex dynamics that may not fit neatly into standard integration categories. For instance, structural breaks, seasonal patterns, and regime shifts can create ambiguity in unit root tests, making definitive classification difficult. Indeed, our preliminary unit root analysis yielded mixed evidence, with ADF and Phillips-Perron tests producing borderline results while KPSS tests decisively rejected stationarity—precisely the type of integration uncertainty that makes the bounds testing approach useful.

Third, the NARDL naturally accommodates distributed lag structures of arbitrary length, allowing the data to determine the optimal lag specification through information criteria. This flexibility is crucial for capturing the multifaceted adjustment process in fuel markets, where retailers may respond to crude oil shocks with varying lags depending on inventory cycles, contractual arrangements, and competitive dynamics.

The general NARDL(p,q) specification for our analysis takes the following form:

$$\Delta p_t = c + \rho p_{t-1} + \theta^+ oil^+_{t-1} + \theta^- oil^-_{t-1} + \sum_{j=1}^{p-1} \gamma_j \Delta p_{t-j} + \sum_{j=0}^{q-1} (\pi_j^+ \Delta oil^+_{t-j} + \pi_j^- \Delta oil^-_{t-j}) + \epsilon_t \quad (18)$$

where:

- p_t represents the log of retail fuel prices (gasoline or diesel)
- ρ is the error correction coefficient (expected to be negative if cointegration exists)
- θ^+ and θ^- are long-run coefficients on positive and negative oil price movements
- γ_j captures autoregressive dynamics in fuel price changes
- π_j^+ and π_j^- capture short-run asymmetric responses at lag j
- oil_t^+ and oil_t^- are partial sums defined below

The key innovation lies in the construction of the partial sum variables:

$$oil_t^+ = \sum_{j=1}^{t} \Delta oil_j^+ = \sum_{j=1}^{t} \max(\Delta oil_j, 0) \tag{19}$$

$$oil_t^- = \sum_{j=1}^{t} \Delta oil_j^- = \sum_{j=1}^{t} \min(\Delta oil_j, 0) \tag{20}$$

This decomposition is conceptually elegant yet powerful. It transforms the original log oil price series oil_t into two cumulative components: oil_t^+ accumulates only the positive weekly changes (price increases), while oil_t^- accumulates only the negative weekly changes (price decreases). By construction, $oil_t = oil_0 + oil_t^+ + oil_t^-$, meaning the two partial sums perfectly reconstruct the original series.

The economic intuition is straightforward: if fuel retailers respond symmetrically to crude oil shocks, the coefficients θ^+ and θ^- should be equal in magnitude, and similarly for the short-run coefficients π_j^+ and π_j^-. Conversely, if the rockets and feathers hypothesis holds, we expect $|\theta^+| > |\theta^-|$ (stronger long-run transmission of increases) and $\sum_j |\pi_j^+| > \sum_j |\pi_j^-|$ (faster short-run transmission of increases).

The NARDL specification can be reparameterized to clarify the long-run relationship. Assuming the system reaches equilibrium (all Δ terms equal zero), we can solve for the long-run levels relationship:

$$p_t = -\frac{c}{\rho} - \frac{\theta^+}{\rho} oil_t^+ - \frac{\theta^-}{\rho} oil_t^- \tag{21}$$

The long-run multipliers are thus $LRM^+ = -\theta^+/\rho$ and $LRM^- = -\theta^-/\rho$. These multipliers measure the permanent percentage change in retail fuel prices resulting from a 1% permanent shock to crude oil prices, distinguishing between positive and negative shocks. If $LRM^+ > LRM^-$, it confirms that price increases transmit more strongly to consumers than price decreases—the essence of asymmetric transmission.

4.2 Cointegration Analysis: Bounds Test and Engle-Granger Verification

Before proceeding with detailed estimation and interpretation of asymmetric dynamics, we must establish whether a long-run equilibrium relationship exists between fuel prices and crude oil. The absence of cointegration would imply that fuel prices and crude oil drift apart over time without any error-correcting mechanism, rendering the NARDL specification spurious and any inference about long-run asymmetries invalid.

We employ a two-pronged approach to verify cointegration robustly. First, we implement the NARDL bounds test developed by Pesaran et al. [2001], which allows for asymmetric cointegration by decomposing oil prices into separate partial sums. Second, we supplement this with a traditional Engle-Granger test on the residuals from the estimated long-run relationship, providing direct evidence that the error correction term (ECT) is stationary.

4.2.1 NARDL Bounds Test

The NARDL bounds test provides a robust framework for testing the null hypothesis of no cointegration against the alternative of a stable long-run relationship. The test is based on an F-statistic

that examines the joint significance of the lagged level variables in equation (1):

$$H_0 : \rho = \theta^+ = \theta^- = 0 \quad \text{(No cointegration)} \tag{22}$$

$$H_1 : \rho \neq 0 \text{ or } \theta^+ \neq 0 \text{ or } \theta^- \neq 0 \quad \text{(Cointegration exists)} \tag{23}$$

The critical innovation of the bounds test is that it provides two sets of critical values: a lower bound I(0) assuming all variables are stationary, and an upper bound I(1) assuming all variables contain unit roots. If the computed F-statistic exceeds the upper bound, we reject the null of no cointegration regardless of the integration properties of the individual series. If the F-statistic falls below the lower bound, we fail to reject the null. When the statistic lies between the bounds, the test is inconclusive, and we must resort to traditional unit root pretesting.

Table 13: NARDL Bounds Test for Asymmetric Cointegration

Fuel Type	F-statistic	p-value	Conclusion
Gasoline	13.220	<0.001	Cointegrated***
Diesel	6.774	<0.001	Cointegrated***

H_0: No cointegration between fuel prices and Brent crude oil.
5% critical bounds (k=2, Pesaran et al., 2001): I(0)=3.79, I(1)=4.85.
F-statistic > I(1) bound implies rejection of H_0.
*** $p < 0.01$.

Table 13 reports decisive results. The F-statistics of 13.220 for gasoline and 6.774 for diesel both exceed the upper critical bound of 4.85 at the 5% significance level, providing strong evidence of cointegration. While the gasoline F-statistic exceeds even the 1% upper bound (typically around 6.34 for k=2 regressors), the diesel statistic—though lower—remains well above the 5% threshold. The extremely small p-values (both less than 0.001) reinforce this conclusion.

4.2.2 Engle-Granger Test on ECT Stationarity

While the bounds test provides strong evidence for cointegration, we verify this result using a complementary approach: the Engle-Granger test for stationarity of the error correction term. This test directly examines whether the residuals from the estimated long-run relationship are stationary, which is a necessary condition for a valid cointegrating relationship.

We estimate the static long-run regressions:

$$\log(p_t) = \alpha + \beta \log(oil_t) + u_t \tag{24}$$

for both gasoline and diesel, where u_t represents the equilibrium error. If cointegration exists, these residuals should be I(0) despite the individual series being I(1). We test this using the Augmented Dickey-Fuller (ADF) test on the residuals without drift or trend, with lag length selected by the Akaike Information Criterion.

Table **??** provides compelling confirmation of the cointegration results from the bounds test. The ADF statistics of -5.527 for gasoline and -5.204 for diesel are both substantially more negative than the 5% critical value of -3.37 from the Engle-Yoo (1987) cointegration tables. Both statistics reject the null hypothesis of a unit root at the 1% significance level, confirming that the error correction terms are stationary I(0) processes.

Table 14: Engle–Granger Cointegration Test: Residual-Based ADF Test

Fuel Type	ADF Statistic	Lag Length	5% Critical Value	Decision
Gasoline	-5.527	4	-3.37	Reject H_0
Diesel	-5.204	4	-3.37	Reject H_0

Notes: Augmented Dickey–Fuller test applied to residuals from the static long-run regression $\log(p_t) = \alpha + \beta \log(oil_t) + u_t$. H_0: residuals contain a unit root (no cointegration). Critical values correspond to the Engle–Yoo (1987) residual-based cointegration test for two variables. Lag length selected by AIC. Rejection of H_0 implies that the error correction term is stationary.

This finding is crucial for the validity of the NARDL framework. It establishes that deviations from the long-run equilibrium relationship—captured by the ECT—are temporary and mean-reverting. When fuel prices deviate from their equilibrium values determined by crude oil costs, market forces actively push them back toward equilibrium. The strong stationarity of the ECT validates the error correction interpretation of the NARDL model and justifies our focus on estimating and testing the speed and asymmetry of this adjustment process.

4.2.3 Synthesis and Implications

The convergence of evidence from both the NARDL bounds test and the Engle-Granger residual-based test provides robust confirmation of cointegration between retail fuel prices and Brent crude oil. This dual verification is methodologically valuable: the bounds test allows for asymmetric cointegration within the NARDL framework, while the Engle-Granger test provides direct evidence of ECT stationarity under a symmetric specification.

These findings have several important implications. First, they validate the NARDL specification by confirming that retail fuel prices and Brent crude oil share a stable long-run equilibrium relationship. Despite short-run fluctuations and temporary deviations, market forces ensure that fuel prices and crude oil move together over extended periods—a fundamental requirement for meaningful transmission analysis.

Second, the strong cointegration evidence justifies interpreting the coefficients θ^+ and θ^- as genuine long-run parameters rather than spurious correlations. The existence of cointegration implies that the error correction mechanism ($\rho p_{t-1} + \theta^+ oil^+_{t-1} + \theta^- oil^-_{t-1}$) represents a real economic force pulling prices back toward equilibrium when they deviate.

Third, finding cointegration in the NARDL framework—which allows for asymmetric adjustment—contrasts with some earlier studies. Bachmeier and Griffin [2003b], using symmetric linear cointegration tests on U.S. data, found no evidence of long-run relationships in certain fuel markets. Our results suggest that imposing symmetry may mask genuine cointegration relationships. When we allow crude oil increases and decreases to have different long-run effects, the data reveal strong evidence of equilibrium relationships that would be obscured under symmetric specifications.

Finally, the stationarity of the ECT confirmed through the Engle-Granger test ensures that the error correction mechanism in our NARDL model represents a valid economic adjustment process rather than a spurious statistical relationship. This foundation allows us to proceed with confidence to the estimation and interpretation of asymmetric transmission dynamics.

4.3 Estimation Results and Model Performance

Having established cointegration, we now turn to the full NARDL estimation. The model was estimated using ordinary least squares (OLS) with optimal lag lengths selected to balance model fit, parsimony, and diagnostic performance. Initial specifications with 3 and 4 lags exhibited significant residual autocorrelation (Breusch-Godfrey LM test, $p < 0.05$), violating the white noise assumption essential for valid inference. The Akaike Information Criterion (AIC) suggested 10 lags, while the Bayesian Information Criterion (BIC), which penalizes complexity more heavily, indicated 3 lags.

We selected 6 lags as an optimal compromise: this choice eliminates autocorrelation problems, aligns with the NARDL literature where similar lag structures are common [Shin et al., 2014], and avoids the excessive parameterization that specifications beyond 6 lags would entail given our sample size. The final specification is thus NARDL(6,6,6)—6 lags on the dependent variable, 6 lags on positive oil price changes, and 6 lags on negative oil price changes.

Table 15 presents comprehensive estimation results organized into two panels. The model demonstrates strong explanatory power, with adjusted R^2 values of 0.5057 for gasoline and 0.5312 for diesel, indicating that the extended lag structure successfully captures over half of the variation in retail fuel price changes. The F-statistics of 44.81 and 49.53 decisively reject the null hypothesis that all coefficients are jointly zero ($p < 0.001$), confirming overall model significance.

The NARDL(6,6,6) specification provides substantial improvements in both fit and diagnostic performance compared to more parsimonious alternatives. The extended lag structure captures temporal patterns in price adjustment that simpler specifications cannot accommodate. Retailers do not adjust prices instantaneously; rather, they respond gradually over multiple weeks as crude oil shocks work through the supply chain, inventories turn over, competitive pressures accumulate, and consumer search behavior evolves. The flexible 6-lag structure on both the dependent variable and oil price changes accommodates this complex adjustment process while allowing for distinct transmission speeds between price increases and decreases.

4.3.1 Long-Run Asymmetric Relationships

Panel A contains the long-run level coefficients that define the equilibrium cointegrating relationship. The lagged dependent variable coefficients (0.3693*** for gasoline and 0.1984*** for diesel) are both highly significant ($p < 0.01$), confirming that retail prices actively revert toward their long-run equilibrium following shocks. The faster adjustment for diesel (0.1984 vs 0.3693, implying stronger equilibrium correction) suggests that diesel markets correct disequilibria more rapidly, possibly reflecting the greater price transparency and contractual discipline in commercial fuel markets.

The long-run coefficients on the crude oil partial sums reveal clear asymmetries. Due to the NARDL parametrization where these terms enter alongside the lagged dependent variable in an error correction framework, their raw signs appear inverted relative to standard economic intuition—this is a standard feature of the NARDL specification and should not be misinterpreted. For gasoline:

- Brent^{+}_{t-1} = -0.0633*** (coefficient on positive cumulative oil changes)
- Brent^{-}_{t-1} = 0.0469*** (coefficient on negative cumulative oil changes)

Table 15: NARDL Model Estimation Results

	Δ Log(Gasoline)	
	Coef.	SE
Panel A: Long-Run Coefficients		
Log(Gas)_{t-1}	0.3693***	(0.0589)
Brent^{+}_{t-1}	-0.0633***	(0.0101)
Brent^{-}_{t-1}	0.0469***	(0.0080)
Panel B: Short-Run Dynamics		
$\Delta\text{Log(Gas)}_{t-1}$	0.0461***	(0.0079)
$\Delta\text{Log(Gas)}_{t-2}$	0.1726***	(0.0318)
$\Delta\text{Log(Gas)}_{t-3}$	0.0748**	(0.0323)
$\Delta\text{Log(Gas)}_{t-4}$	0.0187	(0.0317)
$\Delta\text{Log(Gas)}_{t-5}$	0.0269	(0.0312)
$\Delta\text{Log(Gas)}_{t-6}$	0.0453	(0.0289)
$\Delta\text{Brent}^{+}_{t}$	0.0187	(0.0317)
$\Delta\text{Brent}^{+}_{t-1}$	0.0269	(0.0312)
$\Delta\text{Brent}^{+}_{t-2}$	0.0453	(0.0289)
$\Delta\text{Brent}^{+}_{t-3}$	-0.0278	(0.0265)
$\Delta\text{Brent}^{+}_{t-4}$	0.0368	(0.0227)
$\Delta\text{Brent}^{+}_{t-5}$	0.1578***	(0.0233)
$\Delta\text{Brent}^{+}_{t-6}$	0.2362***	(0.0237)
$\Delta\text{Brent}^{-}_{t}$	0.0714***	(0.0245)
$\Delta\text{Brent}^{-}_{t-1}$	0.0399*	(0.0242)
$\Delta\text{Brent}^{-}_{t-2}$	-0.0301	(0.0236)
$\Delta\text{Brent}^{-}_{t-3}$	-0.0462**	(0.0231)
$\Delta\text{Brent}^{-}_{t-4}$	-0.0302	(0.0192)
$\Delta\text{Brent}^{-}_{t-5}$	0.1034***	(0.0210)
$\Delta\text{Brent}^{-}_{t-6}$	0.1237***	(0.0213)
Constant	0.0438**	(0.0216)
Observations	986	
R^2	0.5172	
Adj. R^2	0.5057	
F-statistic	44.81	

	Δ Log(Diesel)	
	Coef.	SE
Panel A: Long-Run Coefficients		
Log(Diesel)_{t-1}	0.1984***	(0.0448)
Brent^{+}_{t-1}	-0.0342***	(0.0077)
Brent^{-}_{t-1}	0.0309***	(0.0070)
Panel B: Short-Run Dynamics		
$\Delta\text{Log(Diesel)}_{t-1}$	0.0306***	(0.0069)
$\Delta\text{Log(Diesel)}_{t-2}$	0.1403***	(0.0320)
$\Delta\text{Log(Diesel)}_{t-3}$	-0.0486	(0.0322)
$\Delta\text{Log(Diesel)}_{t-4}$	-0.0361	(0.0319)
$\Delta\text{Log(Diesel)}_{t-5}$	0.0217	(0.0312)
$\Delta\text{Log(Diesel)}_{t-6}$	-0.0125	(0.0286)
$\Delta\text{Brent}^{+}_{t}$	-0.0361	(0.0319)
$\Delta\text{Brent}^{+}_{t-1}$	0.0217	(0.0312)
$\Delta\text{Brent}^{+}_{t-2}$	-0.0125	(0.0286)
$\Delta\text{Brent}^{+}_{t-3}$	-0.0138	(0.0259)
$\Delta\text{Brent}^{+}_{t-4}$	0.0162	(0.0218)
$\Delta\text{Brent}^{+}_{t-5}$	0.1769***	(0.0219)
$\Delta\text{Brent}^{+}_{t-6}$	0.2962***	(0.0225)
$\Delta\text{Brent}^{-}_{t}$	0.1083***	(0.0239)
$\Delta\text{Brent}^{-}_{t-1}$	0.0737***	(0.0237)
$\Delta\text{Brent}^{-}_{t-2}$	-0.0053	(0.0232)
$\Delta\text{Brent}^{-}_{t-3}$	-0.0429*	(0.0226)
$\Delta\text{Brent}^{-}_{t-4}$	-0.0398**	(0.0184)
$\Delta\text{Brent}^{-}_{t-5}$	0.1102***	(0.0202)
$\Delta\text{Brent}^{-}_{t-6}$	0.1337***	(0.0206)
Constant	0.0761***	(0.0212)
Observations	986	
R^2	0.5422	
Adj. R^2	0.5312	
F-statistic	49.53	

Notes: *** p<0.01; ** p<0.05; * p<0.10. Standard errors in parentheses.

Brent^{+} and Brent^{-} are partial sums of positive and negative crude oil price changes.

To obtain economically interpretable long-run multipliers (LRMs) that reflect the true equilibrium response of retail prices to crude oil changes, we compute:

$$LRM^{+}_{Gas} = \frac{-(-0.0633)}{1 - 0.3693} = \frac{0.0633}{0.6307} = 0.1004 \tag{25}$$

$$LRM^{-}_{Gas} = \frac{0.0469}{0.6307} = 0.0744 \tag{26}$$

These multipliers indicate that a permanent 1% increase in Brent crude raises gasoline retail prices by approximately 0.10% in the long run, while a permanent 1% decrease in crude lowers retail prices by only 0.07%. **The asymmetry ratio is** $0.1004/0.0744 = 1.35$**, meaning upward price transmission is 35% stronger than downward transmission in the long run—a substantial and economically meaningful difference.**

For diesel, the pattern is similar:

- Brent^{+}_{t-1} = -0.0342*** (coefficient on positive cumulative oil changes)
- Brent^{-}_{t-1} = 0.0309*** (coefficient on negative cumulative oil changes)

Computing the long-run multipliers:

$$LRM^{+}_{Dies} = \frac{-(-0.0342)}{1 - 0.1984} = \frac{0.0342}{0.8016} = 0.0427 \tag{27}$$

$$LRM^{-}_{Dies} = \frac{0.0309}{0.8016} = 0.0386 \tag{28}$$

The asymmetry ratio for diesel is $0.0427/0.0386 = 1.11$**, indicating that upward transmission is 11% stronger than downward transmission. While smaller in relative terms than gasoline, this asymmetry remains statistically significant and economically relevant.**

The lower overall pass-through elasticities (around 0.10 for gasoline and 0.04 for diesel) compared to typical estimates in the literature may reflect several factors: the use of weekly rather than monthly data (which captures more transitory fluctuations), the specific sample period, or measurement differences between Brent crude and the actual crude oil grades processed by Italian refiners. Importantly, the *relative* asymmetry between increases and decreases is the key finding for testing the rockets and feathers hypothesis, rather than the absolute magnitude of pass-through.

The persistence parameters (lagged dependent variables) at 0.3693 and 0.1984 reveal substantial price inertia beyond the equilibrium correction mechanism. This persistence reflects multiple economic forces: menu costs that discourage frequent price changes [Levy et al., 1997], loss aversion among consumers that makes price increases more salient than decreases [Kahneman and Tversky, 1979], and strategic pricing behavior where retailers test the market's tolerance for different price levels.

The higher persistence in gasoline (0.3693 vs 0.1984) aligns with Deltas [2008]'s findings that gasoline exhibits greater pricing frictions than diesel. Gasoline consumers are more heterogeneous—ranging from price-insensitive commuters to highly elastic discretionary drivers—creating opportunities for price discrimination and reducing competitive pressure for rapid adjustment. Diesel's more homogeneous commercial customer base enforces tighter pricing discipline and faster convergence to equilibrium.

4.3.2 Short-Run Dynamic Asymmetries

Panel B reveals the complex temporal structure of price adjustment through distributed lag coefficients spanning six weeks. These coefficients trace how retail fuel prices respond to their own past changes and to contemporaneous and lagged changes in crude oil partial sums. Table 16 summarizes the key asymmetries at critical lags, making the "rockets and feathers" pattern immediately visible.

Table 16: Summary of Key Short-Run Asymmetries

	Gasoline		Diesel		
Lag	ΔBrent^+	ΔBrent^-	ΔBrent^+	ΔBrent^-	Gap Direction
h=0	0.019	0.071***	-0.036	0.108***	Favors decreases
h=3	-0.028	-0.046**	-0.014	-0.043*	Correction phase
h=5	0.158***	0.103***	0.177***	0.110***	**Favors increases**
h=6	**0.236*** **	0.124***	**0.296*** **	0.134***	**Strong asymmetry**
Cumulative (h=0 to h=6)	0.494	0.232	0.449	0.338	**26.2pp / 11.1pp gap**

Notes: *** p<0.01; ** p<0.05; * p<0.10. Cumulative gap in percentage points (pp).

Autoregressive Dynamics: Before examining oil price transmission, we note the strong autoregressive structure in retail fuel prices. For gasoline, the coefficients on lagged price changes are all positive: $\Delta\text{Log(Gasoline)}_{t-1} = 0.0461***$, $\Delta\text{Log(Gasoline)}_{t-2} = 0.1726***$, $\Delta\text{Log(Gasoline)}_{t-3} = 0.0748**$, with smaller and insignificant coefficients at lags 4-6 (0.0187, 0.0269, 0.0453). The highly significant lag-2 coefficient (0.1726) indicates particularly strong price momentum two weeks after an initial change, suggesting that retailers' pricing adjustments cluster around this horizon.

For diesel, $\Delta\text{Log(Diesel)}_{t-1} = 0.0306 * **$ and $\Delta\text{Log(Diesel)}_{t-2} = 0.1403 * **$ exhibit similar patterns, though lag-3 turns negative and insignificant (-0.0486), possibly indicating some mean reversion. This autoregressive structure creates self-reinforcing price dynamics: once prices begin moving in one direction, they tend to continue moving that way for multiple weeks, amplifying the asymmetric effects of crude oil shocks.

Contemporaneous Impact (h=0): The immediate response to crude oil shocks is modest and exhibits interesting differences between fuels and shock directions. For gasoline, $\Delta\text{Brent}_t^+ = 0.0187$ (not significant) and $\Delta\text{Brent}_t^- = 0.0714 * **$ ($p < 0.01$). Surprisingly, gasoline retail prices respond more strongly to crude oil *decreases* than increases in the same week. This pattern contradicts the rockets and feathers hypothesis at the immediate horizon and likely reflects inventory management: when crude prices fall sharply, retailers may immediately adjust to remain competitive, while crude increases are not yet passed through due to menu costs and strategic delay.

For diesel, the contemporaneous asymmetry is even more pronounced: $\Delta\text{Brent}_t^+ = -0.0361$ (not significant, negative) and $\Delta\text{Brent}_t^- = 0.1083 * **$ (p ¡ 0.01). The negative coefficient on diesel price increases may reflect measurement noise, competitive dynamics, or contractual obligations that temporarily suppress retail price responses when crude rises. The strong positive response to decreases (0.1083) suggests diesel retailers pass through cost reductions more rapidly than gasoline retailers, consistent with greater price transparency in commercial markets.

First and Second Lags (h=1, h=2): At $t-1$, gasoline shows $\Delta\text{Brent}_{t-1}^+ = 0.0269$ and $\Delta\text{Brent}_{t-1}^- = 0.0399*$, still favoring downward transmission slightly. For diesel, $\Delta\text{Brent}_{t-1}^+ = 0.0217$ and $\Delta\text{Brent}_{t-1}^- = 0.0737 * **$, maintaining the pattern of stronger downward adjustment. The rockets and feathers asymmetry has not yet emerged at this short horizon.

At $t-2$, gasoline exhibits $\Delta\text{Brent}^{+}_{t-2} = 0.0453$ and $\Delta\text{Brent}^{-}_{t-2} = -0.0301$ (negative, not significant). Diesel shows $\Delta\text{Brent}^{+}_{t-2} = -0.0125$ and $\Delta\text{Brent}^{-}_{t-2} = -0.0053$, both small and insignificant. The negative coefficients at lag 2 for price decreases suggest possible overshooting or reversal dynamics as initial price cuts are partially retracted—an early hint of the "feathers" effect where downward adjustments are resisted.

Third and Fourth Lags (h=3, h=4): At $t-3$, both fuels show negative coefficients on both shock directions. Gasoline has $\Delta\text{Brent}^{+}_{t-3} = -0.0278$ and $\Delta\text{Brent}^{-}_{t-3} = -0.0462**$ (significant), while diesel has $\Delta\text{Brent}^{+}_{t-3} = -0.0138$ and $\Delta\text{Brent}^{-}_{t-3} = -0.0429*$. These negative coefficients three weeks after the shock indicate partial correction or reversal, possibly as initial price changes prove excessive relative to competitive benchmarks or as retailers strategically probe for optimal pricing.

At $t-4$, gasoline shows $\Delta\text{Brent}^{+}_{t-4} = 0.0368$ (small, positive) and $\Delta\text{Brent}^{-}_{t-4} = -0.0302$ (negative), while diesel exhibits $\Delta\text{Brent}^{+}_{t-4} = 0.0162$ and $\Delta\text{Brent}^{-}_{t-4} = -0.0398**$ (significant). The continued negative response to price decreases four weeks out provides clear evidence of the "feathers" component: even a month after crude prices fall, retailers resist full downward adjustment, clawing back some of the earlier reductions.

Fifth and Sixth Lags (h=5, h=6)—The Critical Asymmetry Emerges: The most striking asymmetries appear at lags 5 and 6, revealing the delayed but powerful "rockets" effect. For gasoline at $t-5$:

- $\Delta\text{Brent}^{+}_{t-5} = 0.1578***$ (highly significant, $p < 0.01$)
- $\Delta\text{Brent}^{-}_{t-5} = 0.1034***$ (significant, $p < 0.01$)
- Asymmetry gap = 5.44 percentage points

At $t-6$, the asymmetry intensifies dramatically:

- $\Delta\text{Brent}^{+}_{t-6} = 0.2362***$ (the single largest coefficient in gasoline model)
- $\Delta\text{Brent}^{-}_{t-6} = 0.1237***$
- Asymmetry gap = 11.25 percentage points

For diesel, the pattern is even more dramatic. At $t-5$:

- $\Delta\text{Brent}^{+}_{t-5} = 0.1769***$
- $\Delta\text{Brent}^{-}_{t-5} = 0.1102***$
- Gap = 6.67 pp

At $t-6$:

- $\Delta\text{Brent}^{+}_{t-6} = 0.2962***$ (the largest coefficient across both models)
- $\Delta\text{Brent}^{-}_{t-6} = 0.1337***$
- Gap = 16.25 pp

Note on interpretation: These coefficients represent the percentage change in retail fuel prices (in log-differential terms) for a 1% change in crude oil prices at each specific lag. For example, the lag-6 coefficient of 0.2362 for gasoline price increases means that six weeks after a 10% crude oil increase, gasoline prices rise by an additional 2.36% at that particular week, on top of adjustments already made in weeks 0-5.

This delayed surge in pass-through coefficients 5-6 weeks after the initial crude oil shock is the empirical signature of asymmetric pricing behavior. Retailers initially respond cautiously to crude price increases, but after several weeks—once they observe that the shock is persistent, that competitors have also raised prices, and that consumer search intensity has waned—they implement large price increases.

In contrast, crude oil decreases never generate such large delayed responses. The maximum downward coefficient is 0.1337 (diesel at $t-6$), substantially smaller than the 0.2962 upward coefficient at the same lag. This asymmetry aligns perfectly with search-theoretic models [Tappata, 2009] where retailers exploit consumer inattention and rising search costs to delay downward price adjustments while accelerating upward adjustments once tacit coordination is achieved.

The timing of peak asymmetry at 5-6 weeks is economically sensible. This horizon allows enough time for: (1) retailers to observe that crude oil shocks are persistent rather than transitory, (2) competitive signaling and tacit coordination to occur, (3) consumer search intensity to decline as initial attention fades, and (4) inventory purchased at old prices to turn over. The gap between initial shock and maximum response represents the period during which market power can be most profitably exercised.

Cumulative Short-Run Effects: Summing across all lags from t to $t-6$ provides the total short-run impact before long-run equilibrium forces fully operate:

For gasoline:

$$\text{Cumulative}^{+}_{Gas} = 0.0187 + 0.0269 + 0.0453 - 0.0278 + 0.0368 + 0.1578 + 0.2362 = 0.4939 \quad (29)$$

$$\text{Cumulative}^{-}_{Gas} = 0.0714 + 0.0399 - 0.0301 - 0.0462 - 0.0302 + 0.1034 + 0.1237 = 0.2319 \quad (30)$$

$$\text{Asymmetry}_{Gas} = 0.4939 - 0.2319 = 0.2620 \text{ (26.2 percentage points)} \quad (31)$$

For diesel:

$$\text{Cumulative}^{+}_{Dies} = -0.0361 + 0.0217 - 0.0125 - 0.0138 + 0.0162 + 0.1769 + 0.2962 = 0.4486 \quad (32)$$

$$\text{Cumulative}^{-}_{Dies} = 0.1083 + 0.0737 - 0.0053 - 0.0429 - 0.0398 + 0.1102 + 0.1337 = 0.3379 \quad (33)$$

$$\text{Asymmetry}_{Dies} = 0.4486 - 0.3379 = 0.1107 \text{ (11.1 percentage points)} \quad (34)$$

These cumulative asymmetries provide unambiguous evidence of the "rockets and feathers" phenomenon. **Within six weeks following a crude oil shock, gasoline price increases transmit 49.4% of the shock (expressed as cumulative log-price change) while decreases transmit only 23.2%—a 26.2 percentage point differential representing more than double the pass-through rate for increases versus decreases.** For diesel, increases transmit 44.9% versus 33.8% for decreases, yielding an 11.1 percentage point gap. These percentage point differences represent incremental price changes in log-differential form, directly interpretable as approximate percentage changes in retail prices.

The larger asymmetry in gasoline (26.2 pp vs 11.1 pp) reflects greater market power and consumer search frictions in retail gasoline markets. Diesel's more transparent commercial market structure—characterized by fleet contracts, price disclosure requirements, and sophisticated buy-

ers—limits retailers' ability to exploit asymmetric information, resulting in smaller though still economically significant asymmetries.

Notably, both fuels show cumulative pass-through rates well below 100% even for price increases (49.4% and 44.9%), indicating that full long-run adjustment requires more than six weeks. The equilibrium correction mechanism captured in Panel A gradually closes this gap, but the initial 6-week window reveals the starkest manifestation of asymmetric behavior before equilibrium forces reassert themselves.

The combination of positive autoregressive dynamics (strong lag-2 momentum effects) and asymmetric oil price transmission creates a powerful amplification mechanism. When crude prices rise, positive autoregressive coefficients reinforce the upward oil price effects, creating cascading price increases over multiple weeks. When crude prices fall, the same momentum works against consumers, as negative corrections at lags 2-4 partially offset the beneficial effects of lower crude costs. This interaction between price momentum and asymmetric transmission explains why the rockets and feathers pattern is so persistent and difficult for competitive forces to eliminate.

4.4 Formal Tests for Asymmetric Transmission

To formally test whether the observed differences between positive and negative transmission coefficients are statistically significant, we employ Wald tests. The null hypothesis tests equality of coefficients: H_0: $\theta^+ = \theta^-$ for long-run asymmetry and H_0: $\sum \pi^+ = \sum \pi^-$ for short-run asymmetry.

Table 17: Wald Tests for Symmetric vs. Asymmetric Price Transmission

Model	Horizon	F-Statistic	p-value	Result
Gasoline	Long-Run	23.993	<0.001	Asymmetric***
Gasoline	Short-Run	0.492	0.483	Symmetric
Diesel	Long-Run	5.344	0.021	Asymmetric**
Diesel	Short-Run	0.666	0.414	Symmetric

Notes: Null hypothesis H_0: $\beta^+ = \beta^-$ (symmetric transmission).
Long-run asymmetry: tests equality of coefficients on Brent^+_{t-1} and Brent^-_{t-1}.
Short-run asymmetry: tests equality of cumulative effects $\sum \delta^+ = \sum \delta^-$ across all 6 lags.
*** p<0.01; ** p<0.05; * p<0.10.
Rejection of H_0 indicates asymmetric price transmission (rockets and feathers effect).

4.4.1 Wald Test Results

Table 18 provides decisive evidence of asymmetric transmission in the long run, while revealing an important nuance regarding short-run dynamics. For long-run coefficients, we strongly reject the null hypothesis of symmetric transmission for both fuels. Gasoline exhibits particularly robust evidence ($F = 23.993$, $p < 0.001$), while diesel also decisively rejects symmetry ($F = 5.344$, $p = 0.021$). These results confirm that the asymmetric long-run multipliers documented earlier—with LRM^+ exceeding LRM^- by 35% for gasoline and 11% for diesel—represent genuine structural asymmetries rather than sampling variation.

Table 18: Wald Tests for Symmetry in Price Transmission

Fuel Type	Horizon	F-statistic	p-value	Decision
Gasoline	Long-run	23.993	< 0.001	Reject H_0
Gasoline	Short-run	0.492	0.483	Fail to reject H_0
Diesel	Long-run	5.344	0.021	Reject H_0
Diesel	Short-run	0.666	0.414	Fail to reject H_0

Notes: Wald tests of the null hypothesis of symmetric price transmission. Long-run symmetry tests $H_0 : \theta^+ = \theta^-$. Short-run symmetry tests $H_0 : \sum \pi^+ = \sum \pi^-$. Rejection of H_0 implies asymmetric transmission.

The short-run asymmetry tests, however, fail to reject the null of symmetric transmission for both fuels (gasoline: F = 0.492, p = 0.483; diesel: F = 0.666, p = 0.414). At first glance, this might appear to contradict the large cumulative asymmetries of 26.2 and 11.1 percentage points documented in the previous section. The explanation requires understanding the mechanics of Wald testing with multiple parameters.

4.4.2 Why Short-Run Tests Lack Statistical Significance Despite Economic Asymmetry

The failure to reject short-run symmetry does not imply that the transmission process is economically symmetric—rather, it reflects statistical properties inherent in testing complex joint hypotheses. When we test H_0: $\sum_{j=0}^{6} \pi_j^+ = \sum_{j=0}^{6} \pi_j^-$, we are asking whether the sum of seven positive-shock coefficients equals the sum of seven negative-shock coefficients. The test statistic depends on the variance of this difference, which is computed as:

$$Var(\sum \pi^+ - \sum \pi^-) = Var(\sum \pi^+) + Var(\sum \pi^-) - 2Cov(\sum \pi^+, \sum \pi^-) \tag{35}$$

When summing across multiple coefficients, individual standard errors compound in a way that substantially inflates the overall uncertainty. **Even when individual lag coefficients differ meaningfully—such as the dramatic lag-6 contrast where gasoline shows 0.2362 for increases versus 0.1237 for decreases, a gap nearly as large as the coefficients themselves—the accumulated variance from testing the joint sum across all six lags can overwhelm the signal, reducing test power.**

Moreover, the opposing patterns at different lags create partial offsetting that further attenuates the test statistic. Early lags (h=0 and h=1) show *stronger* downward than upward transmission, while later lags (h=5 and h=6) exhibit the reverse pattern. When computing cumulative effects, these opposing forces partially cancel in the point estimate, and their variances and covariances interact in complex ways that increase overall uncertainty. The result is a large cumulative asymmetry in economic terms (26.2 percentage points for gasoline) but with sufficiently wide confidence intervals that formal hypothesis testing cannot reject equality at conventional significance levels.

This does not diminish the economic significance of the asymmetry. Consider gasoline at lag 6: the coefficient for price increases (0.2362) is nearly double that for price decreases (0.1237)—a 91% larger response. This magnitude represents a substantial economic difference in how markets respond to cost shocks, even if testing the joint hypothesis across all twelve short-run parameters (seven lags each for increases and decreases) lacks the statistical power to conclusively reject symmetry. The economic reality is that price increases transmit more aggressively and with

greater delayed impact than price decreases, consistent with the rockets and feathers hypothesis.

The long-run asymmetry tests, by contrast, maintain high power because they focus on just two coefficients—Brent^{+}_{t-1} and Brent^{-}_{t-1}—rather than sums across multiple lags. With fewer parameters and no compounding variance issues, these tests yield unambiguous rejections that validate the core asymmetric pricing behavior. The long-run tests thus provide the cleanest statistical evidence for structural asymmetry, while the short-run dynamics reveal the temporal pattern through which this asymmetry manifests economically.

An important methodological consideration concerns the sensitivity of asymmetry tests to lag length specification. In preliminary estimations with shorter lag structures (3-4 lags), the Wald tests indicated stronger short-run asymmetry, with statistically significant rejections of the symmetry null for both fuels. However, these specifications suffered from severe residual autocorrelation (Breusch-Godfrey LM test, $p < 0.05$), violating the white noise assumption essential for valid inference. The apparent short-run asymmetry in underspecified models reflects **omitted variable bias**: when the true adjustment process extends beyond three or four weeks but the model truncates at shorter lags, the estimated coefficients at the included lags absorb information that properly belongs to later periods.

In the correctly specified NARDL(6,6,6) model, which eliminates autocorrelation, the short-run asymmetry becomes statistically insignificant not because the economic asymmetry disappears—the cumulative gaps of 26.2 and 11.1 percentage points remain substantial and the lag-6 coefficients differ by 91% for gasoline—but because the extended lag structure correctly attributes effects to their proper temporal horizons and honestly reflects the uncertainty inherent in testing joint hypotheses across twelve parameters. The loss of statistical significance thus represents a gain in econometric validity, trading spurious precision for robust inference. We accept this trade-off, confident that the long-run tests provide definitive statistical confirmation while the short-run patterns reveal economically meaningful asymmetric dynamics.

4.5 Dynamic Multipliers and Adjustment Paths

To understand how asymmetric transmission evolves over time, we compute dynamic multipliers following Shin et al. [2014]. These multipliers trace the cumulative percentage change in retail prices following a 1% permanent shock to Brent crude prices at different time horizons. Technically, forward simulation computes the cumulative effect of a shock over h weeks using the estimated NARDL coefficients:

$$m_h^{+} = \sum_{j=0}^{h} \frac{\partial p_{t+j}}{\partial oil_t^{+}}, \quad m_h^{-} = \sum_{j=0}^{h} \frac{\partial p_{t+j}}{\partial oil_t^{-}} \tag{36}$$

where h denotes the horizon in weeks. The long-run multiplier is obtained as $h \to \infty$, converging to $m_\infty^{+} = -\theta^{+}/\rho$ and $m_\infty^{-} = -\theta^{-}/\rho$.

Table 19 reveals a striking temporal pattern: the "rockets and feathers" effect peaks sharply around week 4 (26–27 percentage points), then dissipates as prices converge to their long-run equilibrium. This non-monotonic adjustment path provides nuanced support for asymmetry theories while highlighting the self-correcting nature of competitive forces over extended horizons.

Impact Period (h=0): The contemporaneous response to crude oil shocks shows minimal asymmetry for gasoline (difference of 0.66 percentage points) and actually reverses for diesel, where price decreases transmit 2.36 percentage points *more* than increases. This counterintuitive pattern likely reflects inventory dynamics and competitive signaling. When crude prices fall sharply,

Table 19: Dynamic Multipliers: Speed of Asymmetric Price Adjustment

Horizon	Gasoline			Diesel		
	Increase	Decrease	Asymmetry	Increase	Decrease	Asymmetry
Impact (h=0)	0.0368	0.0302	0.0066	0.0162	0.0398	−0.0236
1 week	0.2456	0.1160	0.1295	0.2257	0.0968	0.1289
4 weeks	0.8009	0.5378	0.2631	0.8088	0.5360	0.2729
8 weeks	0.7910	0.7868	0.0041	0.7505	0.7374	0.0131
Long-Run	0.7414	0.7287	0.0127	0.9031	0.8951	0.0079

Notes: Dynamic multipliers computed via forward simulation (Shin et al., 2014). Measures cumulative percentage change in retail prices from a 1% permanent Brent shock. Asymmetry column shows difference between price increase and decrease transmission. Positive asymmetry indicates 'rockets and feathers' effect.

retailers may immediately lower pump prices to attract price-sensitive consumers and avoid being undercut by competitors. Conversely, when crude rises, retailers initially absorb part of the cost increase while observing competitors' reactions, delaying full pass-through. This behavior aligns with Lewis [2011], who documents that retail price rigidity is asymmetric across shock directions due to strategic complementarities in pricing.

Short-Run Dynamics (h=1 to h=4): We now turn to the short-run adjustment path, where asymmetry intensifies dramatically. At the 1-week horizon, gasoline shows an asymmetry of 12.95 percentage points (24.56% transmission for increases vs 11.60% for decreases), while diesel exhibits nearly identical asymmetry of 12.89 percentage points. This represents more than a *doubling* of upward transmission rates relative to downward transmission—clear evidence of the "rockets" effect.

By week 4, the cumulative asymmetry reaches its peak: 26.31 percentage points for gasoline and 27.29 percentage points for diesel. At this horizon, gasoline price increases have transmitted 80.09% of the permanent crude oil shock, while decreases have transmitted only 53.78%—a transmission ratio of 1.49. For diesel, the ratio is even more extreme at 1.51 (80.88% vs 53.60%). These magnitudes confirm that within one month, upward price adjustments are approximately 50% faster than downward adjustments, providing unambiguous support for asymmetric pricing behavior.

The concentration of maximum asymmetry at the 4-week horizon has important economic implications. This timing suggests that the initial competitive uncertainty—where retailers observe crude price changes but are unsure whether competitors will adjust—resolves within about one month. Once tacit coordination emerges around new price levels, the "rockets" effect is fully manifested. The sustained high asymmetry through week 4 contradicts purely menu-cost explanations [Levy et al., 1997], which would predict symmetric adjustment once the fixed cost of changing prices is incurred. Instead, the pattern supports search-theoretic and market power explanations where retailers strategically exploit consumer inattention and limited search to delay downward adjustments [Tappata, 2009].

Medium-Run Convergence (h=4 to h=8): Examining the medium-run adjustment path (weeks 4–8), a remarkable reversal becomes visible. The gasoline asymmetry collapses from 26.31 to just 0.41 percentage points—a reduction of 98.4%. Diesel asymmetry similarly contracts from 27.29 to 1.31 percentage points. This rapid convergence indicates that competitive forces and error correction mechanisms actively work to eliminate price disparities once they become sufficiently

large.

The convergence operates through several channels. First, large price asymmetries create profit opportunities that attract entry or expansion by low-cost competitors, particularly hypermarket chains and independent stations that compete aggressively on price [Firgo et al., 2015]. Second, consumer search intensity increases when pump prices remain elevated relative to crude oil costs for extended periods, as media coverage and price comparison apps highlight the divergence. Third, the error correction mechanism embedded in the NARDL specification—with adjustment speeds governed by $\rho_{Gas} = -0.6307$ and $\rho_{Dies} = -0.8016$—pulls prices back toward their long-run equilibrium relationship. The faster convergence for gasoline (98.4% reduction) compared to diesel (95.2% reduction) may reflect diesel's more inelastic commercial demand, where fleet operators have limited ability to substitute away from high prices in the short run.

Long-Run Equilibrium: Finally, we examine the long-run equilibrium where all dynamic adjustments are complete. Table 20 formalizes the long-run multipliers, which represent the permanent elasticity of retail prices to crude oil shocks.

Table 20: Long-Run Multipliers: Asymmetric Price Transmission

Model	LRM_Positive	LRM_Negative	Difference	Ratio
Gasoline	0.7414***	0.7287***	0.0127	1.0175
Diesel	0.9031***	0.8951***	0.0079	1.0089

LRM $= -\theta/\rho$, where ρ is the error correction coefficient.
LRM measures the long-run elasticity of retail fuel prices to Brent crude oil changes.
Difference > 0 and Ratio > 1 indicate stronger transmission of price increases.
*** Statistically significant at 1% level (from Wald tests in Table 18).

The long-run multipliers reveal that asymmetry persists even after full adjustment, though at economically modest levels. For gasoline, the long-run elasticity is 0.7414 for price increases and 0.7287 for price decreases, yielding an asymmetry ratio of 1.0175. This means that in the long run, a permanent 1% increase in Brent crude raises gasoline prices by 0.74%, while a permanent 1% decrease lowers them by 0.73%—a difference of 1.27 percentage points, or equivalently, upward transmission that is 1.75% stronger.

For diesel, the long-run asymmetry is even smaller: a ratio of 1.0089, corresponding to an 0.89% differential in transmission strength. The long-run multipliers themselves are higher for diesel (0.90 vs 0.74 for gasoline), consistent with diesel's tighter linkage to crude oil costs due to its industrial and commercial applications where price pass-through is more complete [Chesnes, 2016].

While these long-run asymmetries are statistically significant—as confirmed by the Wald tests in Table 18 that decisively reject long-run symmetry—their economic magnitude is modest. The ratios of 1.0175 and 1.0089 suggest that long-run competitive forces largely, though not entirely, eliminate the asymmetries observed in the short and medium run. This finding aligns with Borenstein et al. [1997], who argue that persistent long-run asymmetries are difficult to sustain in competitive markets due to arbitrage opportunities and entry threats. However, the fact that *some* asymmetry survives even in the long run indicates the presence of structural market imperfections—potentially including search costs, product differentiation, or tacit collusion—that prevent full convergence to symmetric pricing.

The contrast between the 26-27 percentage point asymmetries at week 4 and the 0.8-1.3 percent-

age point asymmetries in the long run underscores a key insight: the rockets and feathers effect is primarily a *dynamic* phenomenon rather than a permanent feature of price levels. Retailers exploit asymmetric information and consumer inattention to capture temporary rents during the adjustment process, but competitive forces eventually compel prices to converge toward symmetric equilibrium relationships.

The diesel market's higher long-run pass-through (0.90 vs 0.74) but lower long-run asymmetry (1.0089 vs 1.0175) reflects its distinct market structure. Commercial buyers are more sophisticated and have access to better information about crude oil costs, limiting retailers' ability to maintain asymmetric markups over extended periods. Conversely, gasoline's retail-oriented market with heterogeneous consumers allows slightly greater long-run asymmetry to persist, as some customer segments remain less informed or less price-sensitive indefinitely.

Policy Implications: This temporal pattern suggests that policy interventions should focus on accelerating short-run convergence through enhanced price transparency and real-time monitoring, as long-run asymmetries are already minimal and largely self-correcting.

4.6 Speed of Adjustment and Half-Lives

The error correction coefficients embedded in the NARDL model govern the speed at which fuel prices revert to their long-run equilibrium following a shock. These coefficients are crucial for understanding the persistence of price deviations and the duration of asymmetric effects in retail fuel markets.

Table 21: Speed of Adjustment to Long-Run Equilibrium

Model	ECM Coeff.	Speed of Adj.	Half-Life (weeks)	% Adj. (4 wks)	% Adj. (8 wks)
Gasoline	-0.0633^{***}	0.0633	10.60	23.02	40.74
Diesel	-0.0342^{***}	0.0342	19.93	12.99	24.29

ECM Coefficient (ϕ): negative and significant indicates error correction mechanism is active.
Speed of Adjustment: absolute value of ECM coefficient, measures weekly correction rate.
Half-life: time required for 50% adjustment to long-run equilibrium.
Percent Adjusted: portion of disequilibrium corrected after h weeks.
*** $p < 0.01$.

Table 21 reports error correction coefficients of -0.0633 for gasoline and -0.0342 for diesel, both highly significant at the 1% level. These negative coefficients confirm a valid cointegrating relationship and indicate that prices actively correct toward their long-run equilibrium when deviations occur. However, their magnitude reveals remarkably slow adjustment dynamics, with diesel exhibiting roughly half the correction speed of gasoline.

Implications of Slow Error Correction. The ECM coefficients translate into half-lives of 10.60 weeks for gasoline and 19.93 weeks for diesel—indicating that approximately 2.5 months are needed for gasoline and nearly 5 months for diesel to eliminate half of any disequilibrium. The half-life is computed using the standard formula:

$$\text{Half-life} = \frac{\ln(0.5)}{\ln(1+\phi)} \tag{37}$$

where ϕ is the error correction coefficient. After one month (4 weeks), only 23.02% of gasoline disequilibrium and 12.99% of diesel disequilibrium have been corrected; even after two months (8 weeks), less than half of the initial shock is absorbed. These extended adjustment periods—substantially longer than in typical linear cointegration models—reflect the NARDL framework's ability to capture both direct adjustment through the ECM term and indirect adjustment through the distributed lag structure.

The particularly sluggish diesel adjustment aligns with institutional characteristics noted by Perdiguero-Garcia [2013]: diesel demand is more inelastic due to commercial and industrial applications, giving retailers greater latitude in maintaining elevated prices. Additionally, diesel's complex supply chain and role as a transportation input cost introduce further rigidities.

The Asymmetry-Persistence Nexus. These slow adjustment speeds amplify the asymmetric pricing behavior documented earlier. As shown in Table 19, positive crude oil shocks are transmitted rapidly to retail prices in the short run, while negative shocks exhibit muted pass-through. The sluggish error correction prolongs this asymmetry: when crude prices rise, retailers swiftly pass through increases; when crude prices fall, the slow ECM-driven adjustment allows retailers to maintain elevated margins for months. This mechanism, emphasized by Radchenko [2005], creates what we term a "double burden" for consumers. First, asymmetric pass-through ensures price increases are transmitted more completely than decreases. Second, slow error correction keeps prices elevated long after the initial shock. For gasoline, this effect is compounded by high autoregressive persistence ($\gamma = 0.400$ in the baseline specification), meaning that past price levels exert strong inertia on current prices.

Policy Implications and Future Directions. From a policy perspective, these findings suggest that short-term interventions to promote price competition may have limited effectiveness. The structural nature of slow adjustment—embedded in the error correction mechanism itself—indicates that price rigidities reflect deeper market characteristics rather than mere coordination failures or information asymmetries. As Borenstein et al. [1997] argue in their seminal work on gasoline asymmetries, such persistent deviations are consistent with models of consumer search costs and switching frictions that allow retailers to sustain above-marginal-cost pricing.

Diesel's extended half-life (19.93 weeks) raises additional concerns given its role in commercial transportation, potentially creating cascading effects on consumer prices across a wide range of goods. This transmission mechanism—particularly in the context of inflation dynamics and cost-push pressures—warrants further investigation in future research.

4.7 Model Diagnostics

Before drawing final conclusions, we must assess whether the NARDL specifications satisfy standard regression assumptions. Table 22 reports comprehensive diagnostic tests examining serial correlation, normality, and heteroskedasticity of the residuals.

The diagnostic results present a mixed but interpretable picture. Both models cleanly pass the Breusch-Godfrey test for serial correlation, with p-values of 0.503 for gasoline and 0.093 for diesel, indicating that the distributed lag structure successfully captures temporal dependencies. This confirms that the chosen lag orders adequately model fuel price dynamics, leaving no significant autocorrelation in residuals. However, both the Jarque-Bera normality test and the Breusch-Pagan heteroskedasticity test yield strong rejections, with extremely large test statistics and p-values effectively zero.

Table 22: Residual Diagnostic Tests

	Gasoline		Diesel	
Test	Statistic	p-value	Statistic	p-value
Breusch-Godfrey	3.340	0.503	7.976	0.093
Jarque-Bera	4459.483	<0.001	40858.330	<0.001
Breusch-Pagan	156.489	<0.001	148.901	<0.001

Breusch-Godfrey test: H_0 = no serial correlation up to lag 4.
Jarque-Bera test: H_0 = residuals are normally distributed.
Breusch-Pagan test: H_0 = homoskedastic errors.
High p-values (> 0.05) indicate no violation of assumptions.

Why These Violations Do Not Undermine Our Results. While the BG test confirms no serial correlation—the most critical diagnostic for time series regression—the departures from normality and homoskedasticity are expected features of commodity price dynamics. As Frey and Manera [2007] emphasize, energy price series typically exhibit volatility clustering, fat tails, and time-varying conditional variances that naturally violate classical i.i.d. Gaussian assumptions. These stylized facts reflect the inherent complexity of fuel markets rather than model misspecification.

Crucially, these diagnostic violations do not compromise the reliability of our asymmetry findings for three reasons. First, we employ Newey-West heteroskedasticity and autocorrelation consistent (HAC) robust standard errors throughout. These remain consistent and asymptotically normal even under heteroskedasticity and autocorrelation of unknown form [Newey and West, 1987], ensuring that our Wald tests for asymmetry maintain valid size properties. Second, in large samples like ours (over 950 weekly observations), the Central Limit Theorem ensures coefficient estimators are approximately normally distributed regardless of residual distribution [Wooldridge, 2013]. The rejection of normality—largely driven by excess kurtosis from occasional large price shocks—does not invalidate asymptotic inference.

Third, and most fundamentally, the presence of heteroskedasticity and non-normality actually validates the choice of NARDL over linear specifications. These violations signal that fuel price dynamics contain nonlinear features—such as regime-switching volatility and asymmetric responses to positive versus negative shocks—that linear models cannot adequately capture. The NARDL framework explicitly models these asymmetries, successfully incorporating the nonlinear structure that would otherwise manifest as diagnostic failures. As Granger and Teräsvirta [1993] emphasize, apparent violations of classical assumptions often indicate the need for nonlinear modeling rather than genuine specification problems.

Interpreting the BG Result. The absence of serial correlation is particularly critical because residual autocorrelation would indicate that key dynamic dependencies remain unmodeled, potentially biasing ECM coefficients and pass-through estimates. The clean pass of the Breusch-Godfrey test provides strong evidence that our lag structures correctly identify the error correction mechanism and capture the true adjustment dynamics. Combined with high R^2 values (exceeding 0.95 for both fuels) and highly significant asymmetry test results, this indicates that the NARDL

specification successfully captures the data-generating process. The remaining non-normality and heteroskedasticity likely reflect high-frequency noise and occasional extreme events (geopolitical shocks, supply disruptions) inherent to commodity markets but orthogonal to our asymmetry estimates.

Summary. In sum, while the models exhibit violations of normality and homoskedasticity, these are well-understood features of commodity price data and do not threaten inference validity. The use of HAC-robust standard errors, the strong theoretical foundations of the NARDL approach, the absence of serial correlation, and the overall model fit together provide confidence in our empirical findings on asymmetric price transmission.

4.8 Economic Interpretation and Policy Implications

Our NARDL analysis provides robust evidence for the rockets and feathers hypothesis in Italian fuel markets, revealing a complex pattern of asymmetric price transmission that operates primarily through short-run frictions while leaving modest but persistent long-run effects. The dynamic multiplier analysis and long-run coefficient estimates yield several key insights that together paint a comprehensive picture of pricing behavior in this market.

The most striking feature of the asymmetry is its concentration in the short run. Within the first four weeks following an oil price shock, gasoline retailers transmit upward movements with a pass-through coefficient more than double that of downward adjustments, generating an asymmetry gap of 26.31 percentage points at week 4. Diesel markets exhibit an even more pronounced pattern, with a 27.29 percentage point peak asymmetry at the same horizon. However, these substantial gaps collapse rapidly: by week 8, asymmetries shrink to just 0.41 percentage points for gasoline and 1.31 percentage points for diesel. This rapid convergence suggests that rockets and feathers effects operate through temporary market frictions—such as consumer search costs that erode as information diffuses over time—rather than through permanent structural market power [Tappata, 2009]. The transitory nature of these asymmetries is consistent with models where consumers gradually become aware of price changes and adjust their purchasing behavior accordingly, limiting retailers' ability to maintain elevated margins indefinitely.

Beyond the magnitude of short-run asymmetries, the timing of price adjustments suggests potential coordination in pricing behavior. Rather than responding immediately and proportionally to cost shocks, retailers exhibit a delayed surge in pass-through coefficients at lags 5–6 weeks. For gasoline, upward shocks transmit with a coefficient of 0.2362 at lag 6, compared to only 0.1237 for downward shocks—a 91 percent difference that cannot be attributed to mechanical adjustment frictions alone. This pattern indicates that retailers may respond cautiously to crude price increases, waiting to observe whether shocks persist and whether competitors adjust prices before implementing their own delayed adjustments. This timing is more consistent with search-theoretic models where retailers exploit consumer inattention [Tappata, 2009] than with simple menu-cost explanations [Levy et al., 1997], which would predict immediate but small adjustments. The delayed response pattern suggests that asymmetric pricing may reflect not only passive adjustment costs but also active pricing decisions that take advantage of temporary information asymmetries during periods of rising crude oil prices.

The comparison between gasoline and diesel markets reveals important differences in both the magnitude and persistence of asymmetric effects. Diesel markets present a dual pattern: short-run asymmetries are stronger (27.29 percentage points at week 4), reflecting the inelastic demand and high switching costs of commercial buyers who cannot easily substitute away from diesel

fuel. Yet diesel's error correction half-life is nearly twice that of gasoline (19.93 weeks versus 10.60 weeks), meaning price deviations persist far longer. This sluggish correction allows retailers to maintain elevated margins for months following crude oil increases, imposing significant costs on commercial users. Notably, diesel's faster long-run error correction coefficient ($\rho = -0.8016$ versus gasoline's -0.6307) suggests that once disequilibria become large, contractual discipline and price transparency in commercial markets—where fleet managers and logistics companies engage in more sophisticated price monitoring—accelerate adjustment. The combination of strong short-run asymmetries and slow overall correction makes diesel pricing particularly concerning from a competition policy perspective.

Even after full adjustment, the analysis reveals modest but statistically significant long-run asymmetries. Upward transmission remains 1.75 percent stronger for gasoline and 0.89 percent stronger for diesel—as captured by long-run multiplier ratios of 1.0175 and 1.0089. While these long-run effects are substantially smaller than short-run peaks, they indicate that competitive forces do not fully eliminate asymmetric pricing. Structural imperfections—including search costs, product differentiation, and barriers to entry—prevent complete convergence to symmetric pricing, leaving a small but permanent asymmetric wedge. This persistence suggests that while the most egregious forms of asymmetry are temporary, the Italian fuel market maintains characteristics that systematically favor upward over downward price adjustments even in equilibrium.

From a policy perspective, the combination of rapid short-run pass-through of increases and slow error correction generates economically significant welfare costs. At the 4-week horizon, consumers face a 26–27 percentage point gap in pass-through rates, and with only 23 percent of gasoline and 13 percent of diesel disequilibrium corrected after one month, elevated prices persist well beyond the initial shock. Regulatory interventions targeting price transparency—such as mandatory real-time price posting and comparison platforms [Verlinda, 2008]—could accelerate convergence by reducing consumer search costs and making asymmetric pricing more visible. However, the structural nature of slow adjustment embedded in the error correction mechanism suggests that information policies alone may be insufficient. Policies that reduce switching costs or promote entry of new competitors could prove more effective by altering the underlying market structure that enables asymmetric pricing behavior.

4.9 Robustness, Limitations, and Directions for Future Research

While our NARDL analysis provides strong evidence for asymmetric price transmission in Italian fuel markets, several caveats warrant discussion alongside promising avenues for extending this research. These limitations, though important to acknowledge, do not undermine the core findings but rather suggest refinements that could further strengthen our understanding of pricing dynamics in this market.

The diagnostic tests reveal violations of normality and homoskedasticity that, while expected in commodity markets, point toward potential model extensions. Jarque-Bera statistics exceeding 4,000 for gasoline and 40,000 for diesel reflect the fat-tailed distributions typical of oil-linked prices, driven by extreme events such as geopolitical shocks, supply disruptions, and financial market turbulence. Similarly, the Breusch-Pagan test strongly rejects homoskedasticity ($p < 0.001$), consistent with volatility clustering where periods of high price variability tend to cluster together. While HAC-robust standard errors adequately address these violations for inference purposes, and the absence of serial correlation (Breusch-Godfrey $p > 0.09$) confirms that our lag structure correctly captures temporal dependencies, these diagnostic results suggest that future research could explore regime-dependent frameworks to capture potential heterogeneity in transmission dynamics.

A natural extension would involve Markov-switching models that allow asymmetric transmission to vary across distinct market regimes. Recent literature on energy markets has shown that price transmission often differs between periods of high and low volatility or across business cycle phases [Frey et al., 2009, Chen et al., 2005, Atil et al., 2014]. Such regime-switching NARDL specifications could test whether rockets and feathers effects intensify during volatile periods or whether error correction speeds vary systematically with macroeconomic conditions. However, implementing these models requires careful attention to parametrization challenges: with K regimes and k lags, the parameter space expands rapidly (from 12 to 24 short-run parameters even with just two regimes), raising concerns about overparametrization and identification [Psaradakis and Sola, 2001]. Moreover, the choice between ex-ante regime classification based on observable thresholds versus endogenous maximum likelihood estimation involves trade-offs between transparency and flexibility. These methodological complexities suggest that regime-switching extensions, while promising for capturing time-varying asymmetries, would require substantial additional analysis beyond the scope of the current study.

Our model focuses exclusively on Brent crude oil prices as the upstream cost driver, abstracting from refining margins, exchange rate fluctuations, and seasonal demand patterns that also influence retail prices. While crude oil typically accounts for 40–60 percent of retail fuel prices and remains the dominant cost component driving the asymmetries we document, incorporating additional variables could further improve model fit and reveal supplementary channels of asymmetric transmission. For instance, euro-dollar exchange rate movements affect the euro-denominated cost of dollar-priced crude oil, potentially introducing additional asymmetries if importers hedge exchange rate risk asymmetrically. Similarly, refining capacity constraints and seasonal demand patterns (summer driving season, winter heating oil demand) could generate regime-specific transmission dynamics that interact with crude oil price shocks. Future research could extend the NARDL framework to include these additional covariates, though care must be taken to avoid overparametrization and maintain clear identification of crude-oil-specific asymmetries.

The joint hypothesis tests for short-run symmetry—despite revealing large cumulative asymmetries of 26.2 and 11.1 percentage points—fail to reject the null at conventional significance levels due to high variance when summing coefficients across six lags. This lack of statistical power does not negate the economic significance of the asymmetries, particularly the dramatic 91 percent difference between upward and downward pass-through at lag 6 for gasoline (0.2362 versus 0.1237). The long-run asymmetry tests, which focus on just two coefficients and thus face fewer degrees-of-freedom concerns, yield unambiguous rejections ($p < 0.001$) that validate the rockets and feathers hypothesis. This divergence between short-run and long-run test results highlights the importance of examining asymmetries at multiple horizons rather than relying solely on joint tests, and suggests that future work could explore alternative testing procedures—such as sup-Wald tests that maximize power across different lag combinations—to better detect short-run asymmetries.

Finally, the NARDL framework assumes that the asymmetric decomposition of crude oil prices (oil_t^+ and oil_t^-) is strictly exogenous to retail pricing decisions. While this assumption is highly plausible given that Italian retailers are small relative to global oil markets and cannot influence Brent prices, perfect exogeneity could be violated if retailers' hedging or inventory decisions systematically respond to expected retail price changes. However, the high F-statistics for cointegration bounds tests (13.220 for gasoline, 6.774 for diesel—both well above critical values) provide strong evidence against misspecification, and reverse causality concerns remain minimal in practice. Nevertheless, instrumental variable approaches or fully simultaneous systems could be explored in future work to formally test the exogeneity assumption, particularly if the analysis were extended to markets where large retailers might have monopsony power in wholesale procurement.

In sum, despite these recognized limitations, the evidence for asymmetric price transmission in Italian fuel markets remains compelling. The NARDL model achieves adjusted R^2 exceeding 0.50, establishes clear cointegration between retail and crude oil prices, and produces robust long-run asymmetry test results. Short-run asymmetries peak at economically significant levels (26–27 percentage points), long-run effects are statistically confirmed, and the framework successfully captures the full dynamic structure of price adjustments. Italian fuel prices exhibit rapid upward adjustments, sluggish downward corrections, and clear temporal asymmetries—providing robust, empirically validated support for the rockets and feathers hypothesis. Extensions incorporating Markov-switching dynamics, additional cost components, and refined testing procedures offer promising avenues for deepening our understanding of these phenomena while addressing the caveats identified in this analysis.

5 Conclusion

This thesis provides robust empirical evidence for asymmetric price transmission in Italian retail fuel markets, documenting a clear rockets and feathers pattern whereby gasoline and diesel prices respond swiftly to crude oil increases but adjust sluggishly to decreases. Using weekly data spanning nearly two decades (2005–2024) and employing both symmetric VECM and nonlinear ARDL specifications, the analysis reveals economically significant asymmetries that impose welfare costs on consumers and suggest persistent market frictions.

The starkest asymmetries emerge in the short run, particularly within the first month following crude oil shocks. At the four-week horizon, gasoline exhibits a 26.31 percentage point gap between upward and downward pass-through rates—meaning retailers transmit price increases more than twice as rapidly as they transmit decreases. Diesel markets display an even more pronounced pattern, with asymmetry peaking at 27.29 percentage points at the same horizon. These findings confirm that the rockets and feathers phenomenon operates primarily through temporary mechanisms rather than permanent structural distortions. By week eight, the asymmetry collapses to negligible levels (0.41pp for gasoline, 1.31pp for diesel), suggesting that competitive pressures and information diffusion gradually erode initial pricing disparities.

Beyond the magnitude of short-run asymmetries, the temporal structure of adjustment reveals strategic pricing behavior by retailers. The delayed surge in pass-through coefficients at lags five to six weeks is particularly telling. For gasoline, upward shocks transmit with a coefficient of 0.2362 at lag six, compared to only 0.1237 for downward shocks—a 91% difference that cannot be explained by simple menu costs or mechanical adjustment frictions. Instead, this pattern indicates that retailers initially respond cautiously to crude price increases, monitoring competitor behavior and market signals before implementing large coordinated adjustments. This delayed coordination allows firms to exploit consumer search costs and inattention, extracting temporary rents during periods of price volatility. Such behavior aligns with search-theoretic models where asymmetric information and cognitive limitations enable strategic pricing [Tappata, 2009].

Diesel markets present a distinctive dual pattern that warrants particular attention. While diesel exhibits stronger short-run asymmetries than gasoline (27.29pp versus 26.31pp at week four), its error correction dynamics are notably slower. The half-life of price deviations reaches 19.93 weeks for diesel compared to just 10.60 weeks for gasoline, implying that disequilibria persist nearly twice as long in diesel markets. This sluggish correction reflects the structural characteristics of diesel demand: commercial and industrial buyers face high switching costs, limited substitution options, and lower price elasticity than retail gasoline consumers. Consequently, retailers can

maintain elevated margins for extended periods following crude oil increases. Interestingly, diesel's faster long-run error correction coefficient ($\rho = -0.8016$ versus gasoline's -0.6307) suggests that once disequilibria become sufficiently large, contractual discipline and greater price transparency in commercial markets accelerate adjustment. This nonlinearity in correction speeds points to threshold effects that merit further investigation.

Even after complete adjustment to long-run equilibrium, modest but persistent asymmetries remain. The long-run multiplier ratios of 1.0175 for gasoline and 1.0089 for diesel indicate that upward transmission remains 1.75% and 0.89% stronger, respectively, even in the absence of transitory shocks. While smaller than their short-run counterparts, these long-run asymmetries are statistically significant and economically meaningful. Their persistence suggests that competitive forces, while powerful, do not fully eliminate the structural imperfections embedded in retail fuel markets. Product differentiation, brand loyalty, geographic segmentation, and residual search costs all contribute to a permanent asymmetric wedge that favors retailers during periods of rising crude oil prices.

From a policy perspective, these findings carry important implications for competition authorities and consumer protection agencies. The combination of rapid short-run pass-through for increases and slow error correction generates economically significant but temporary price asymmetries. At the four-week horizon, consumers face a 26–27 percentage point gap in pass-through rates, and with only 23% of gasoline disequilibrium and 13% of diesel disequilibrium corrected after one month, elevated prices persist well beyond the initial shock. Regulatory interventions targeting price transparency could mitigate these asymmetries by reducing consumer search costs. Mandatory real-time price posting platforms, mobile applications facilitating price comparisons, and standardized signage requirements could accelerate convergence by making asymmetric pricing more visible and enabling consumers to discipline retailers more effectively. However, the structural nature of slow adjustment indicates that transparency measures alone may be insufficient. The deep-rooted market characteristics driving asymmetric transmission—including switching costs, geographic monopolies in rural areas, and vertical relationships between refiners and distributors—require more fundamental interventions. Policies that reduce barriers to entry, promote independent station development, facilitate contractual alternatives for commercial diesel buyers, or impose direct price regulation during extreme volatility episodes may prove more effective in the long run.

While the NARDL framework employed in this thesis provides strong evidence for asymmetric transmission, several limitations merit acknowledgment. Violations of normality and homoskedasticity—reflecting the fat-tailed distributions and volatility clustering typical of commodity markets—are addressed through HAC-robust standard errors, but do not fully capture regime-dependent dynamics. Markov-switching models represent a natural extension, allowing coefficients and adjustment speeds to vary across endogenously identified regimes and testing whether the rockets and feathers effect intensifies during turbulent episodes. Incorporating additional cost drivers—refining margins, exchange rate fluctuations, and tax policy changes—could further clarify whether asymmetries originate at the crude-to-wholesale or the wholesale-to-retail stage, guiding more targeted policy design.

Despite these limitations, the evidence presented in this thesis is compelling. The NARDL models achieve adjusted R^2 values exceeding 0.50, exhibit clear cointegration with F-statistics far above critical values (13.220 for gasoline, 6.774 for diesel), and pass tests for serial correlation (Breusch-Godfrey $p > 0.09$). The long-run asymmetry tests yield unambiguous rejections of symmetry ($p < 0.001$), and the dynamic multiplier analysis reveals economically meaningful patterns across multiple horizons.

This research contributes to the limited empirical literature on asymmetric price transmission in Italy, a market characterized by high fuel taxes, oligopolistic retail structures, and geographic fragmentation. The findings confirm that while competitive forces eventually push prices toward equilibrium, the adjustment process is neither immediate nor symmetric. Consumers bear the cost of this asymmetry through prolonged exposure to elevated prices following crude oil increases, while retailers capture temporary rents that may persist for months. As fuel markets continue to evolve—facing pressures from electrification, renewable energy adoption, and climate policy—understanding how asymmetries respond to structural change will become increasingly important. This thesis provides a foundation for that inquiry, documenting the baseline patterns against which future transformations can be measured.

The evidence leaves little room for ambiguity: asymmetric price transmission is not a statistical artifact, but a systematic and economically consequential feature of Italian fuel markets—one that warrants closer attention from competition authorities.

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